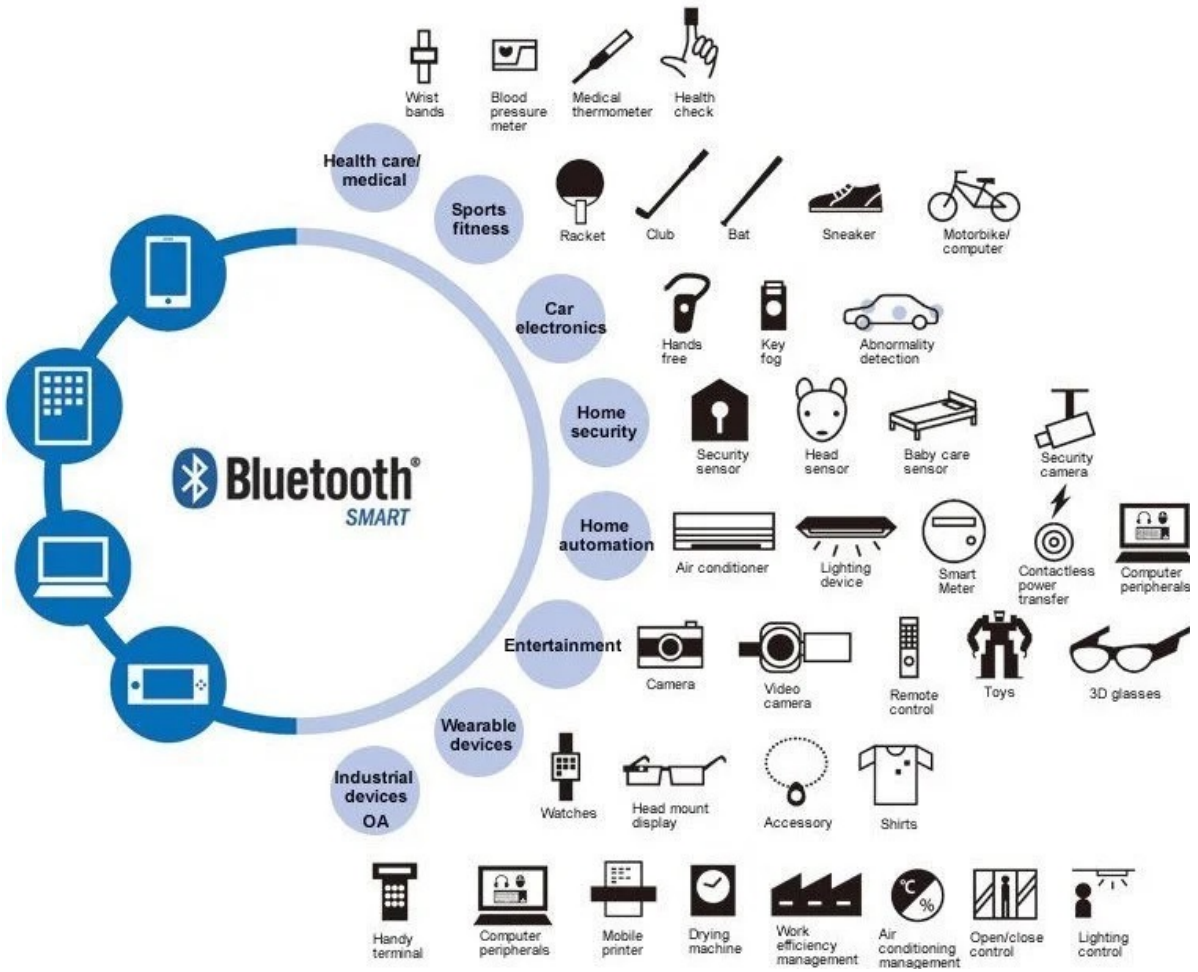
Wireless Communication Network Standards

for the Internet of Things (IoT)

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CAMINHOS QUE CONECTAM COM O FUTURO
PATHWAYS THAT CONNECT US TO THE FUTURE

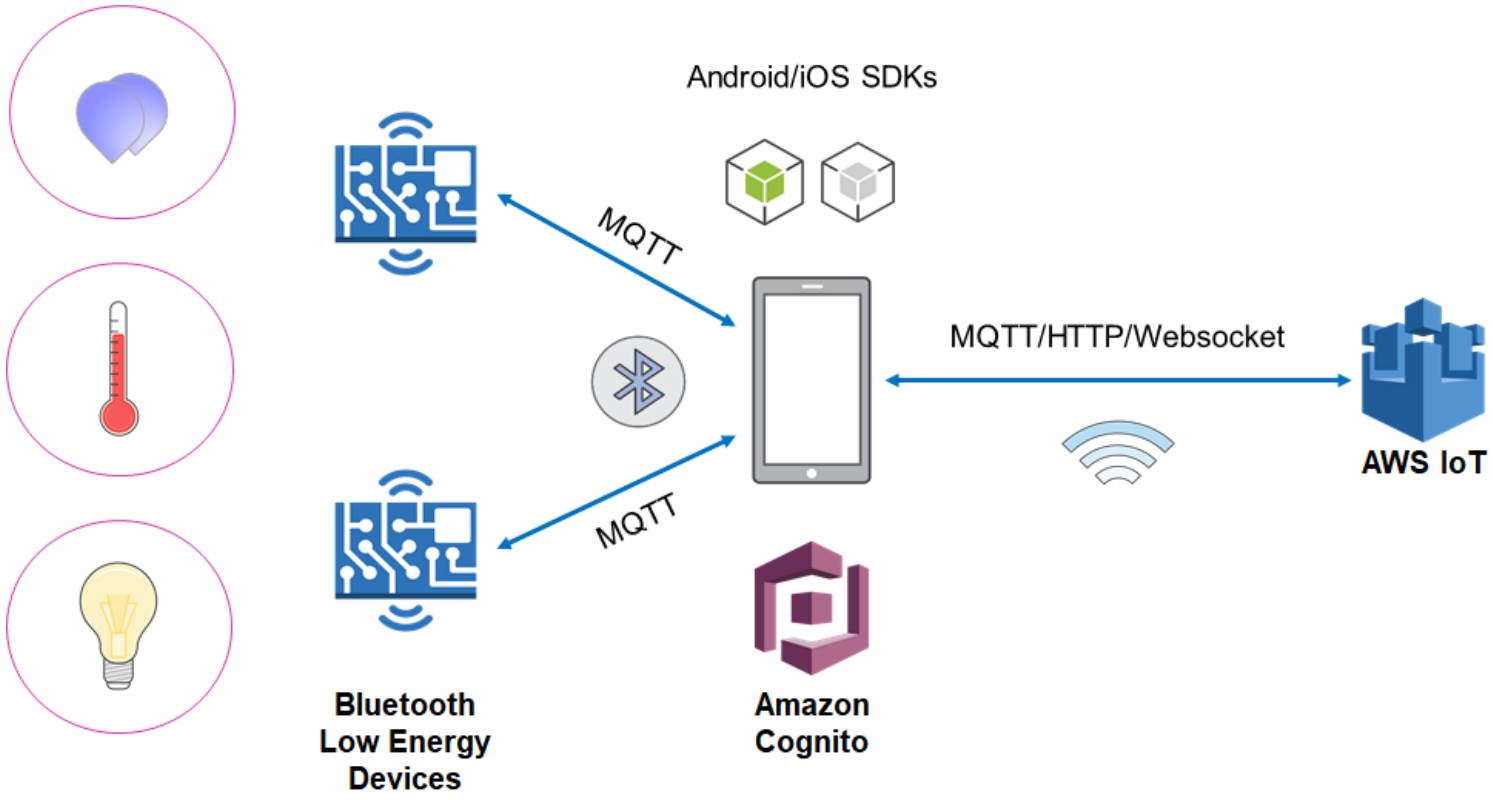
BLUETOOTH LOW ENERGY



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IEEE 802.15.1 (BLUETOOTH)



BLUETOOTH LOW ENERGY

- The Bluetooth Low Energy (LE) radio is designed for very low-power operation. It transmits data over 40 channels in the unlicensed 2.4 GHz ISM band.
- The Bluetooth LE radio gives developers considerable flexibility to build products that meet the specific connectivity requirements of their markets.
- Bluetooth LE supports several communication topologies, from point-to-point and broadcast to, more recently, mesh. This allows Bluetooth to support reliable, large-scale device networks.

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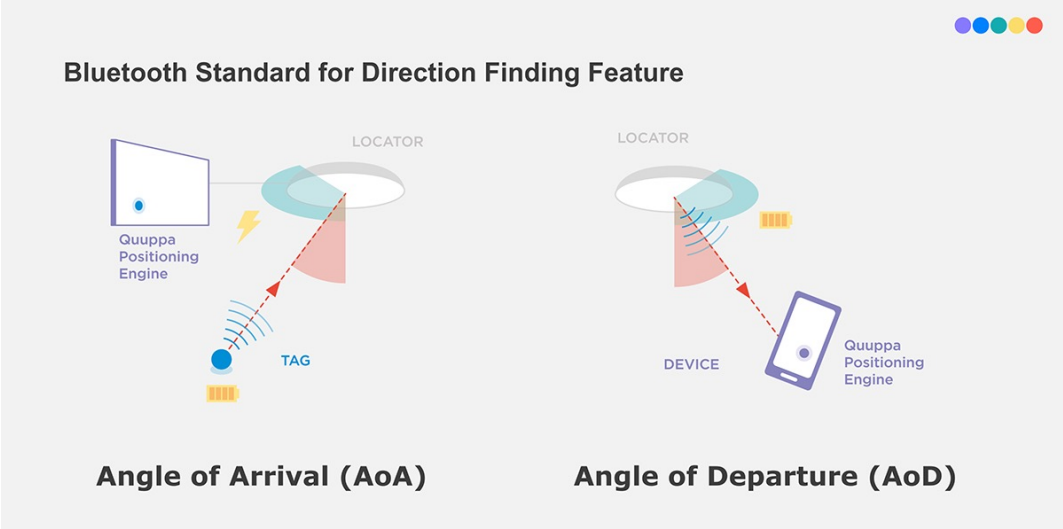
BLUETOOTH LOW ENERGY

- BLE has a wide range of uses, including healthcare, fitness, security, home automation, home entertainment, smart industry, and the Internet of Things (IoT). It is also readily available in the smartphones and laptops we use every day.



BLUETOOTH LOW ENERGY

- Initially known for device communication, Bluetooth LE is now also widely used for device positioning to meet the growing demand for high-accuracy indoor location services.
- Originally supporting simple presence and proximity features, Bluetooth LE now supports Bluetooth® Direction Finding.



BLUETOOTH LOW ENERGY

	Bluetooth Low Energy (LE)	Bluetooth Classic
Frequency Band	2.4GHz ISM Band (2.402 – 2.480 GHz Utilized)	2.4GHz ISM Band (2.402 – 2.480 GHz Utilized)
Channels	40 channels with 2 MHz spacing (3 advertising channels/37 data channels)	79 channels with 1 MHz spacing
Channel Usage	Frequency-Hopping Spread Spectrum (FHSS)	Frequency-Hopping Spread Spectrum (FHSS)
Modulation	GFSK	GFSK, π/4 DQPSK, 8DPSK
Data Rate	LE 2M PHY: 2 Mb/s LE 1M PHY: 1 Mb/s LE Coded PHY (S=2): 500 Kb/s LE Coded PHY (S=8): 125 Kb/s	EDR PHY (8DPSK): 3 Mb/s EDR PHY (π/4 DQPSK): 2 Mb/s BR PHY (GFSK): 1 Mb/s
Tx Power*	≤ 100 mW (+20 dBm)	≤ 100 mW (+20 dBm)
Rx Sensitivity	LE 2M PHY: ≤-70 dBm LE 1M PHY: ≤-70 dBm LE Coded PHY (S=2): ≤-75 dBm LE Coded PHY (S=8): ≤-82 dBm	≤-70 dBm
Data Transports	Asynchronous Connection-oriented Isochronous Connection-oriented Asynchronous Connectionless Synchronous Connectionless Isochronous Connectionless	Asynchronous Connection-oriented Synchronous Connection-oriented
Communication Topologies	Point-to-Point (including piconet) Broadcast Mesh	Point-to-Point (including piconet)
Positioning Features	Presence (Advertising) Proximity (RSSI) Direction (AoA/AoD) Distance (Coming)	None

* Devices shall not exceed the maximum allowed transmit power levels set by the regulatory bodies that have jurisdiction over the locales in which the device is to be sold or intended to operate. Implementers should be aware that the maximum transmit power level permitted under a given set of regulations might not be the same for all modulation modes.

BLUETOOTH LOW ENERGY

- A BLE device acts in a central or peripheral role and is sometimes called a client or a server.
- Central (client). A device that initiates commands and requests and accepts responses. Examples include computers and smartphones.
- Peripheral (server). A device that receives commands and requests and returns responses. Examples include a temperature sensor and a heart rate monitor.

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BLUETOOTH LOW ENERGY

- Headphones, fitness trackers, and heart rate monitors are typical BLE peripherals. A device acts as a peripheral when it advertises its availability for a connection and provides a communication interface.
- A device that discovers, connects to, and interacts with peripherals acts as a central, as a computer or smartphone might. A central often scans for nearby peripherals and lists the devices it finds. If they are connectable, it can connect to a selected peripheral and start exchanging information.

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BLUETOOTH LOW ENERGY

- BLE uses a hierarchical data structure to define how information is exchanged. A BLE peripheral advertises services and characteristics that can be used to communicate with it. These attributes are defined using a Generic Attribute Profile (GATT).
- Characteristics expose values as small pieces of information that can change over time. Characteristics of similar types are grouped into services.

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BLUETOOTH LOW ENERGY

- Characteristics can hold readable or writable values. Peripherals with readable characteristics generally publish information for the central to use.
- Peripherals with writable characteristics generally provide an interface for receiving data from a central, such as a configuration value that changes something on the peripheral.

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BLUETOOTH LOW ENERGY

- Suppose a peripheral uses BLE to provide a temperature reading from a sensor.
- It would have a readable temperature characteristic containing the current temperature value.
- As the temperature readings change, the peripheral updates the characteristic value.
- A connected central, such as a computer or smartphone, can then subscribe to changes in that characteristic to receive real-time temperature readings over the peripheral's BLE connection.

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- Each characteristic and service defined for a GATT profile has a unique identifier (UUID).
- The Bluetooth SIG defines some standard characteristic types as reserved. BLE peripherals must uniquely identify their characteristics.

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BLUETOOTH LOW ENERGY

- A central device typically follows this sequence:
  - Scan for peripherals advertising BLE service types identified by UUID.
  - Connect to a selected peripheral.
  - Discover services on the connected peripheral.
  - Discover characteristics of the services of interest on the peripheral.
  - Read or write values for the characteristics of interest.
  - Disconnect when the connection is no longer needed.

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BLUETOOTH LOW ENERGY

- BLE sends and receives data in small packets. BLE 4.0 and 4.1 define a maximum data size of 27 bytes.
- The packet also contains a 4-byte header, for a total size of 31 bytes.
- Given the limits on packet size, it is important to consider how to structure data for BLE transmission.
- Developing for BLE involves converting values to binary representations that can be serialized and deserialized consistently at both ends.

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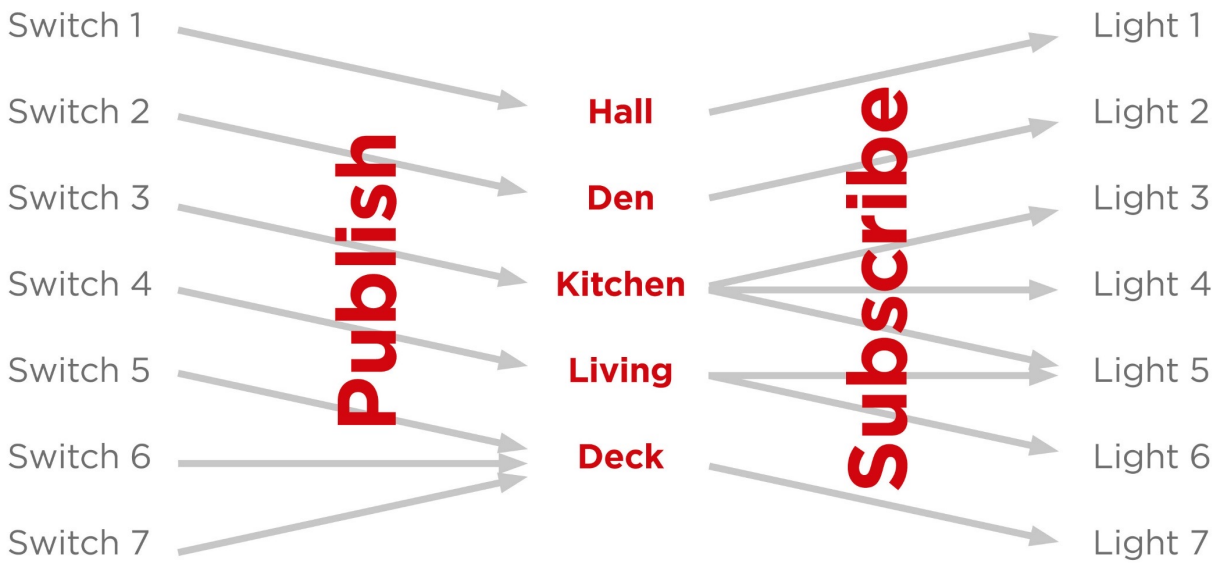
BLUETOOTH MESH

- Bluetooth Mesh provides a framework for large-scale industrial deployments using technology that is already secure, reliable, and supported by all smartphones.
- Bluetooth Mesh builds on Bluetooth to create a mesh network. Each device can communicate with any other nearby device within range, without having to communicate with a central hub.



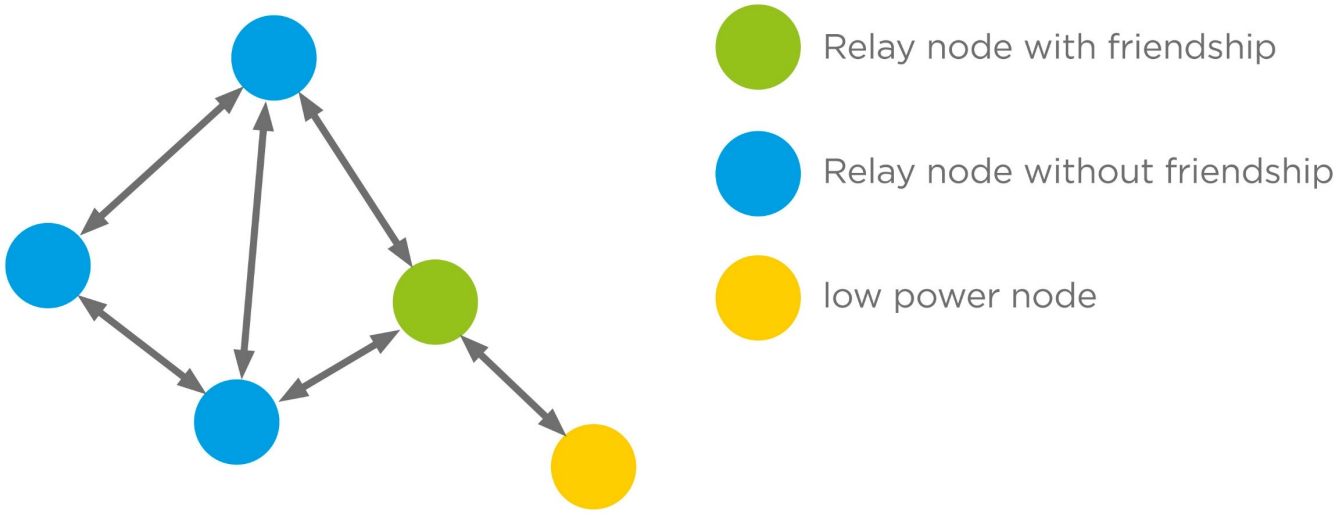
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BLUETOOTH MESH





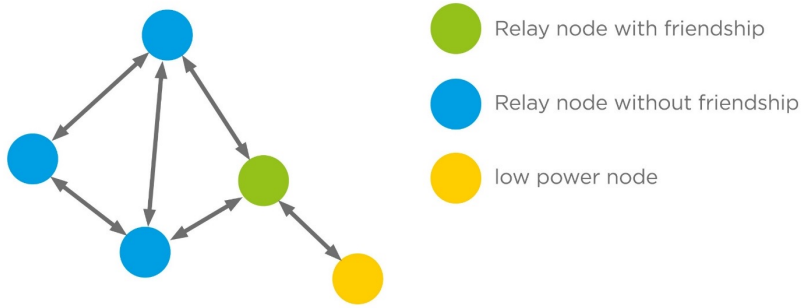
BLUETOOTH MESH



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BLUETOOTH MESH

- Bluetooth Mesh offers two main benefits. First, it extends network range. Each device only needs to be within 20–30 meters of another device, instead of within 20–30 meters of a network hub.
- Second, it improves network resilience. With no single point of failure, data can be routed around a node that goes offline.



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BLUETOOTH MESH

- Warehousing is an area where the Industrial Internet of Things (IIoT) can bring significant gains. A mesh network makes it easier to use beacons to track assets, so you know exactly where your equipment is and whether it is in use.



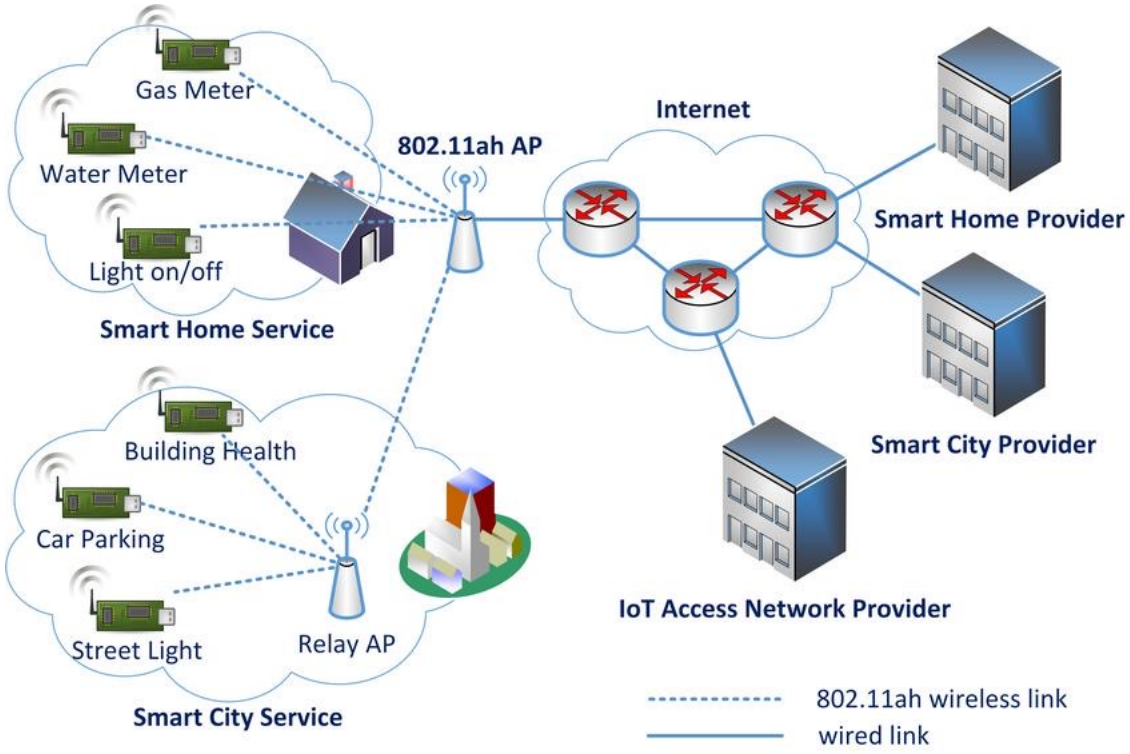


BLUETOOTH MESH

- Connected machinery can also report performance data in real time, allowing predictive maintenance to address problems before they become failures.
- A mesh network gives you much greater control over the environment. For example, you can attach humidity sensors to each product, which is important for highly sensitive materials, and connect them to the HVAC system so all products are stored under the right conditions.

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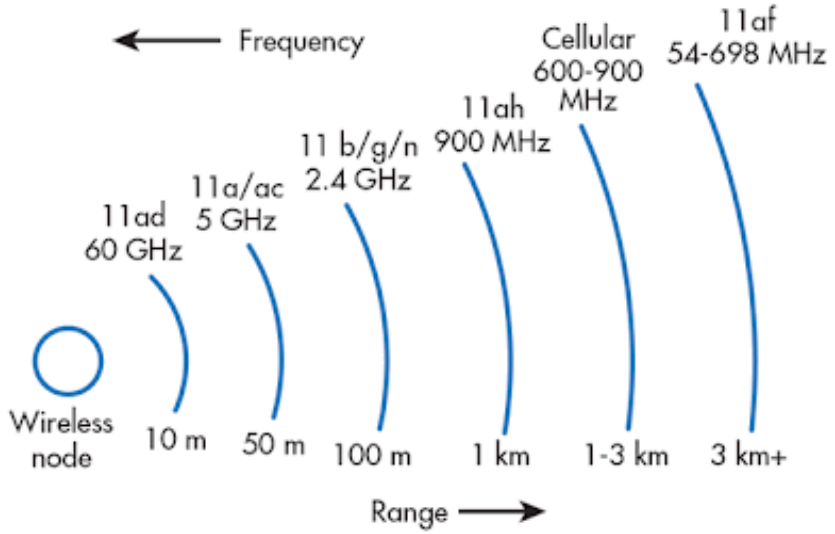
IEEE 802.11.ah



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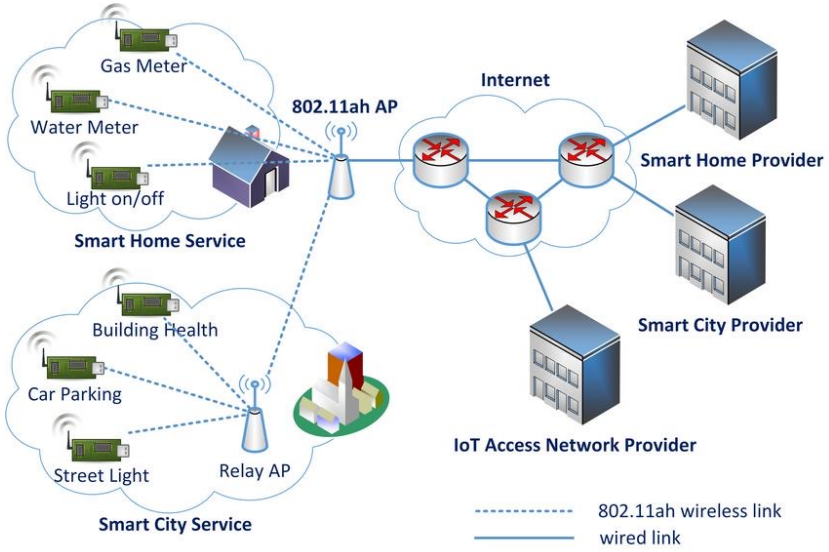
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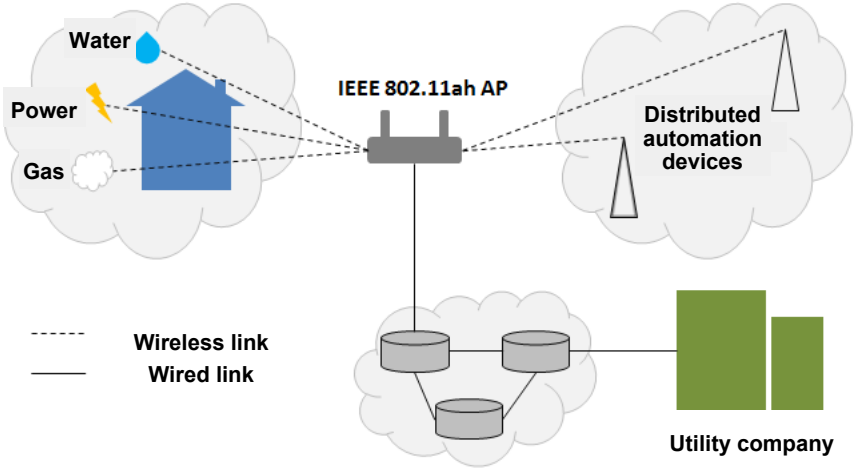
- Developed by the IEEE 802.11 group.
- Wireless networks operating at frequencies below 1 GHz.
- Longer-range wireless networks.
- Support for more devices.



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IEEE 802.11.ah



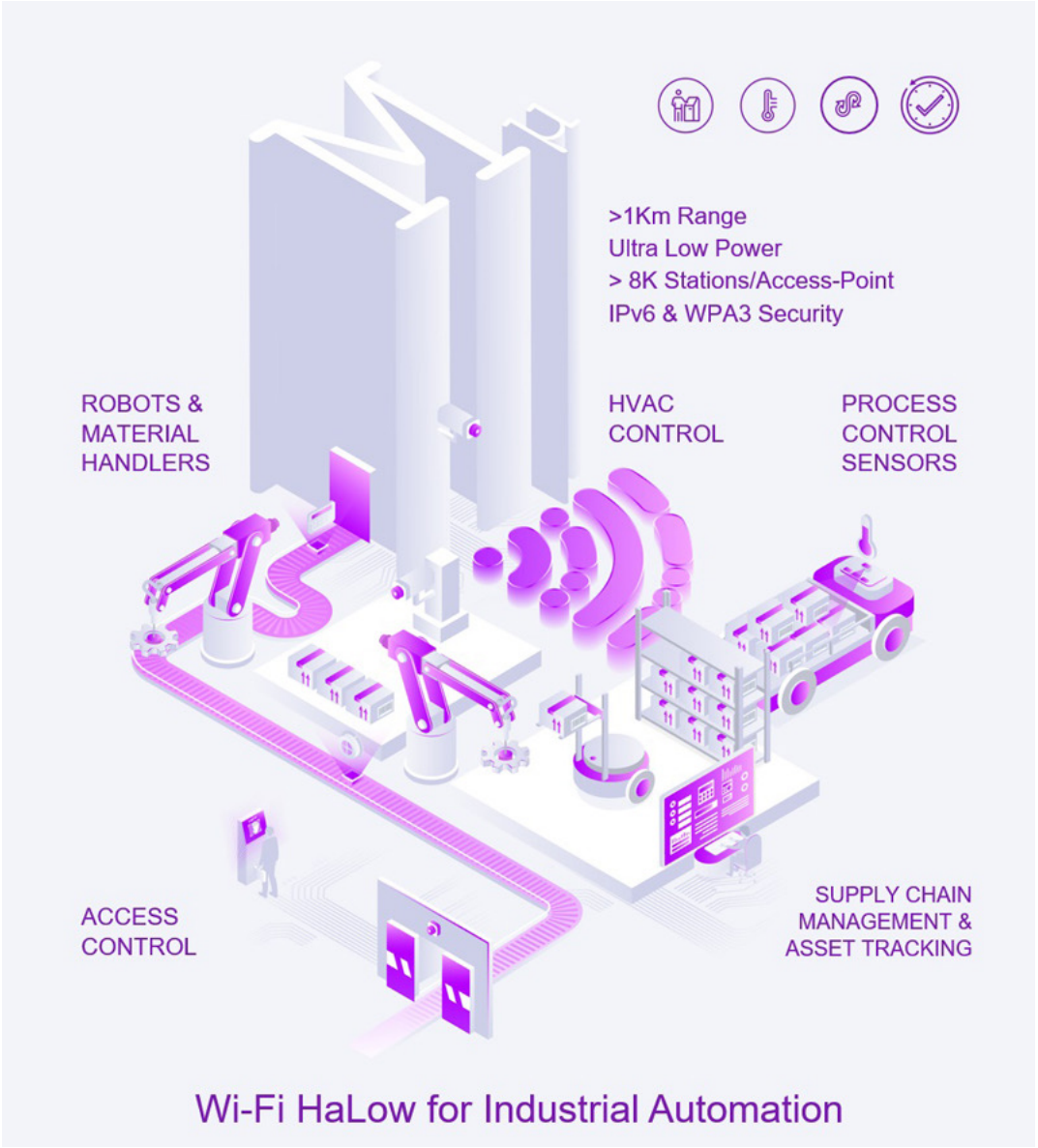




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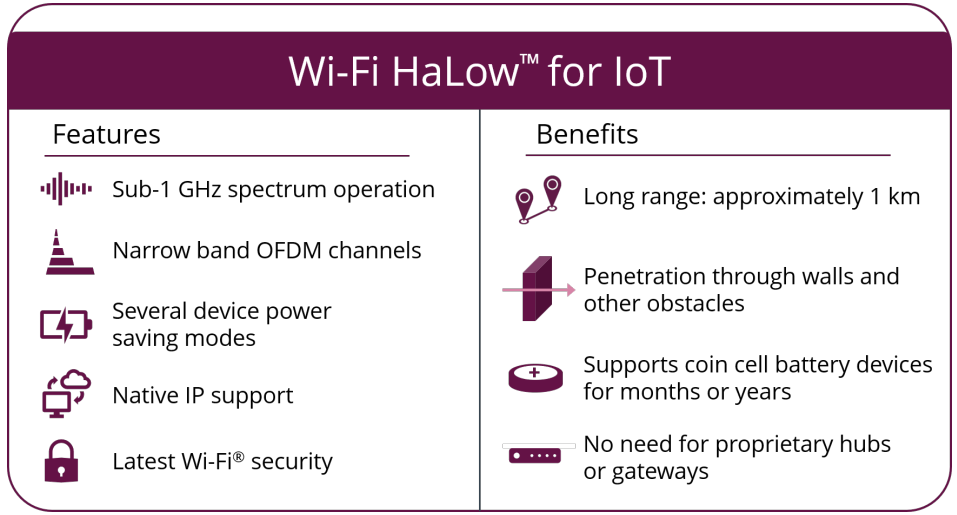
IEEE 802.11.ah



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IEEE 802.11.ah

- IEEE 802.11ah offers several advantages:
  - Ease of use in outdoor environments.
  - Excellent propagation at low frequencies.
  - High sensitivity.
  - Long battery life and power-saving strategies.



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IEEE 802.11.ah

	802.11ac use cases and requirements [3, 4]	802.11ah IoT use case and requirements [2]
Use cases	1) Wireless display 2) Distribution of HDTV 3) Rapid upload/download	Sensors and meters
Data rate requirement	20 Mb/s–3 Gb/s	100 kb/s
Single frame packet size	Large (e.g., 1500 bytes)	Small (e.g., few 100 bytes)
Traffic type	Video streaming/large file transfer	Periodic packet transmisson every few to tens minutes
Distance between devices	5–60 m	Up to 1 km
Number of stations	3–20	Up to 6000
Location	Mostly indoor	Indoor and outdoor

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IEEE 802.11.ah

- Given the diversity and complexity of IoT applications and environments, the next Wi-Fi standard needs to meet these requirements to support M2M applications:
  - Up to 8,191 devices associated with one access point (AP).
  - Carrier frequencies below 1 GHz.
  - Transmission range of up to 1 km outdoors.
  - Data rates of at least 100 kbps.
  - Short, infrequent data transmissions.
  - Very low power consumption.
  - A cost-effective solution.

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IEEE 802.11.ad

Millimeter-wave communication



IEEE 802.11.ad

- With a large block of spectrum in the 60 GHz band available worldwide for unlicensed use, new technologies are emerging that enable Wi-Fi communication in this frequency band.



IEEE 802.11.ad

- Data rates of several gigabits per second are needed to transmit signals such as uncompressed video.
- One example is unencoded high-definition 3D video streams, currently at 4K resolution (Ultra HD, 3840 × 2160 pixels), with a trend toward higher frame rates such as 48 frames per second.
- The 802.11ad amendment to the WLAN standard defines MAC and PHY layers for very high throughput (VHT) in the 60 GHz band.

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IEEE 802.11.ad

- Why use the 60 GHz band?
- An unlicensed frequency band in the 60 GHz range is available worldwide.
- This range allows wider channels and higher throughput.
- Another advantage is the short wavelength (approximately 5 mm). It makes compact, competitive antennas or antenna arrays possible, for example for beamforming.
- At first glance, this range also has some apparent disadvantages:
  - Very high free-space attenuation (68 dB at 1 m and 91 dB at 10 m).
  - Oxygen (O_2) absorption.

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IEEE 802.11.ad

- Since transmission typically occurs over a limited range of less than 10 m, as in a typical living room, high attenuation can also be an advantage.
  - Interference from nearby transmissions is very unlikely.
  - The transmission is very difficult to intercept, which makes it more secure.

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IEEE 802.11.ad

- The most significant difference at 60 GHz is the increase in signal attenuation.
- The network is designed for exchanging information within a single room or space.

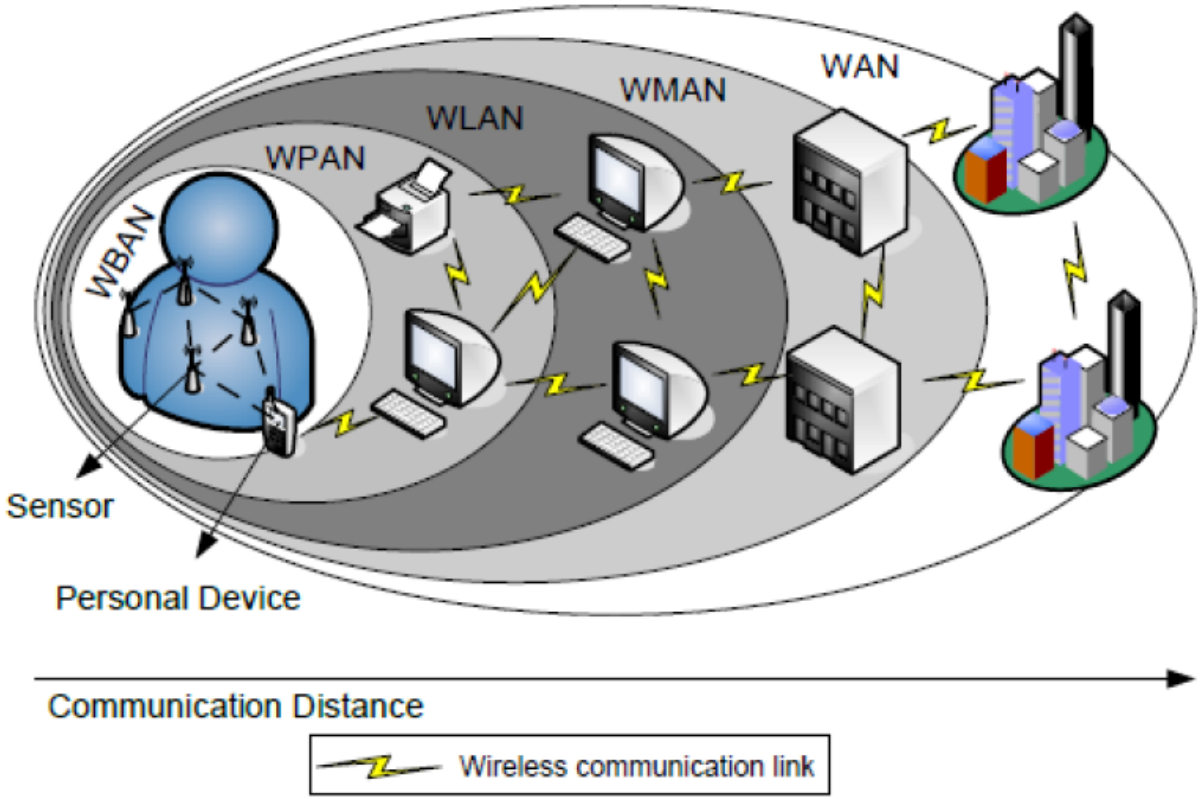


IEEE 802.11.ad

- Typical applications for 802.11ad include:
  - Wireless displays.
  - Distribution of HDTV content, for example in living rooms.
  - Wireless PC connections for rapidly transferring very large files.
  - Automatic synchronization applications.

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WIRELESS BODY NETWORKS



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Wireless Body Networks

- Wireless body area networks (WBANs) emerged as a key technology for real-time patient health monitoring and for diagnosing many life-threatening diseases.
- IEEE 802 established the IEEE 802.15.6 Task Group to standardize WBANs.
- The group aims to establish a communication standard optimized for low-power nodes inside and outside the body, for a variety of medical and nonmedical applications.
- It also aims to define new physical and medium access control layers for WBANs.

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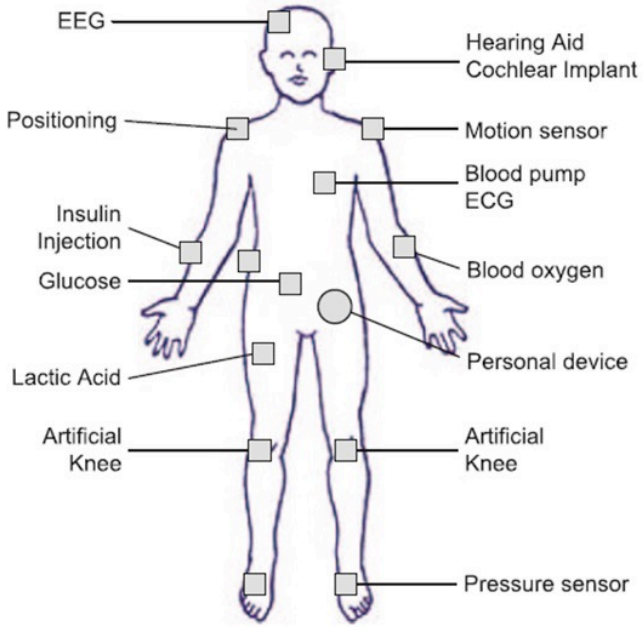
Wireless Body Networks

- Released in 2012.
- Aims to establish an international standard for wireless communication systems with these characteristics:
  - Low power consumption.
  - High reliability.
  - Short-range operation.
  - Use near or inside the human body.

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Wireless Body Networks

- The ideal locations for monitoring different vital signs, applying electrical stimulation, or injecting drugs differ.
- A WBAN must:
  - Provide communication links on or around the body.
  - Enable communication between sensors, actuators, and elements that process power and information.

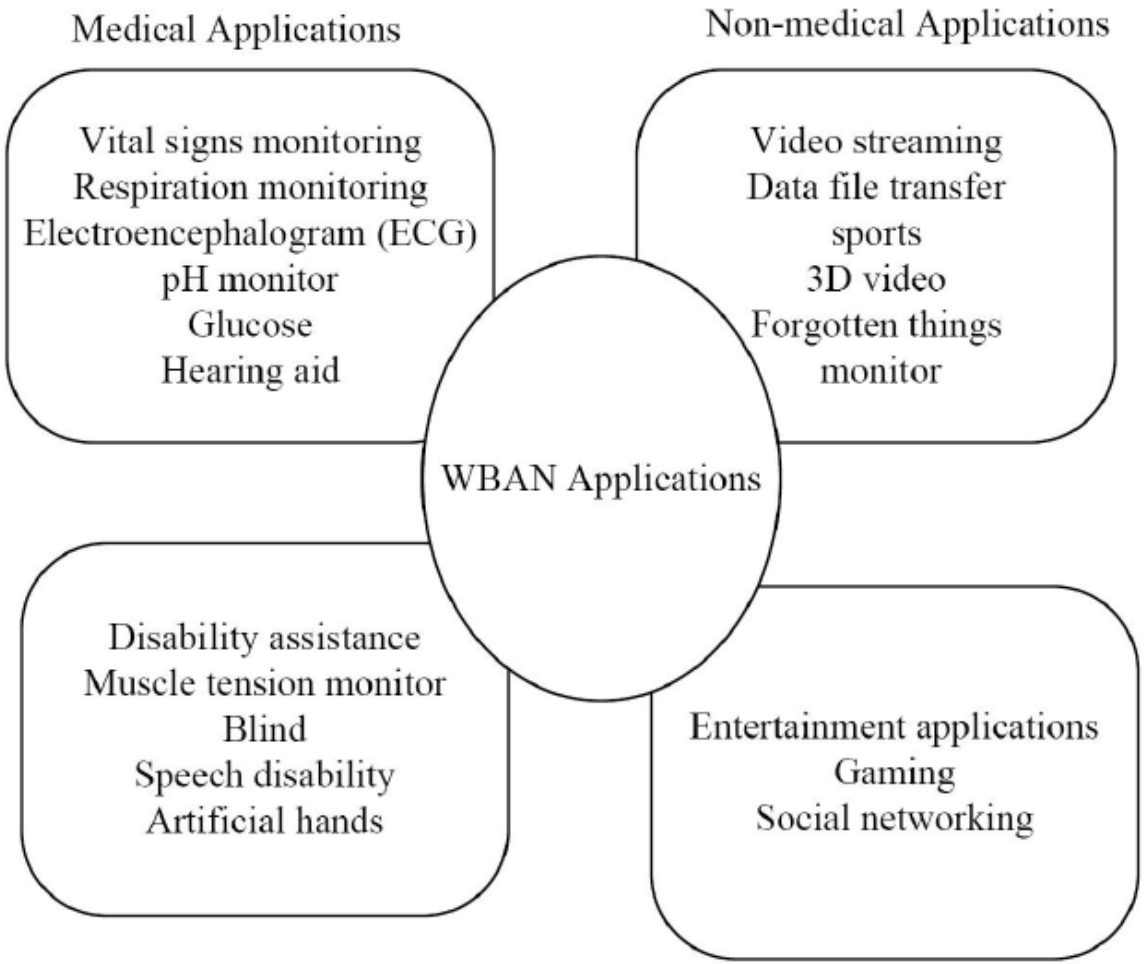


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Wireless Body Networks

Sensor	How it works	Data rate
Accelerometer	Measures the acceleration relative to freefall in three axes	High
Gyroscope	Measures the orientation, based on the principles of angular momentum	High
ECG/EEG/EMG	Measures potential difference across electrodes put on corresponding parts of the body	High
Pulse oximetry	Measures ratio of changing absorbance of the red and infrared light passing from one side to the other of a thin part of the body's anatomy	Low
Respiration	Uses two electrodes, cathode and anode covered by a thin membrane to measure the oxygen dissolved in a liquid	Low
Carbon dioxide	Uses the infrared light and measures the absorption of the gas presented	Low
Blood pressure	Measures the systolic pressure (peak pressure) and diastolic pressure (minimum pressure)	Low
Blood sugar	Traditionally analyzes drops of blood from a finger tip, recently, uses non-invasive method including a near infrared spectroscopy, ultrasound, optical measurement at the eye, and the use of breath analysis	Low
Humidity	Measures the conductivity changes of the level of humidity	Very low
Temperature	Uses a silicon integrated circuit to detect the temperature changes by measuring the resistance	Very low

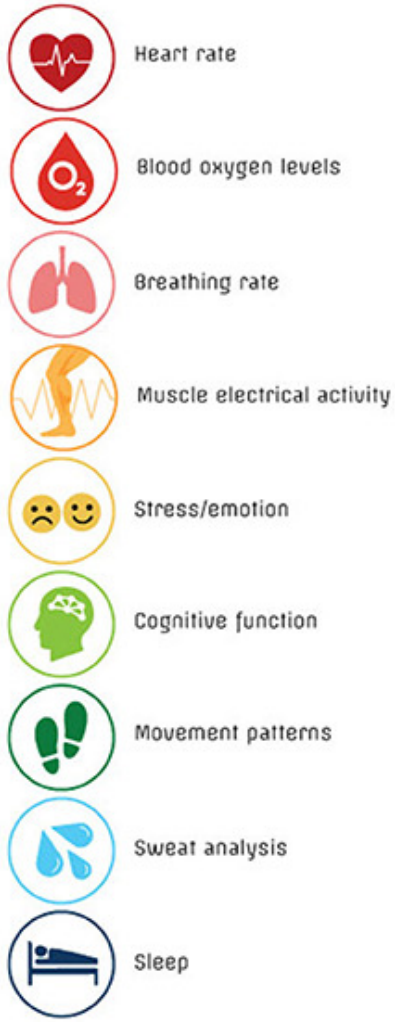
WIRELESS BODY NETWORKS



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Wireless Body Networks





Wireless Body Networks

WBANs consist of several heterogeneous devices that communicate with one another over a wireless network around the human body. These devices can communicate with wearable applications for human health monitoring. WBAN device types include:

1. Wireless sensor nodes respond to physical stimuli, collect and process data, and report the information wirelessly.
2. Wireless actuator nodes act on data from controller nodes, which may originate from external commands or user interaction.
3. Controller nodes aggregate information collected by sensor nodes and send commands to actuator nodes.

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Wireless Body Networks

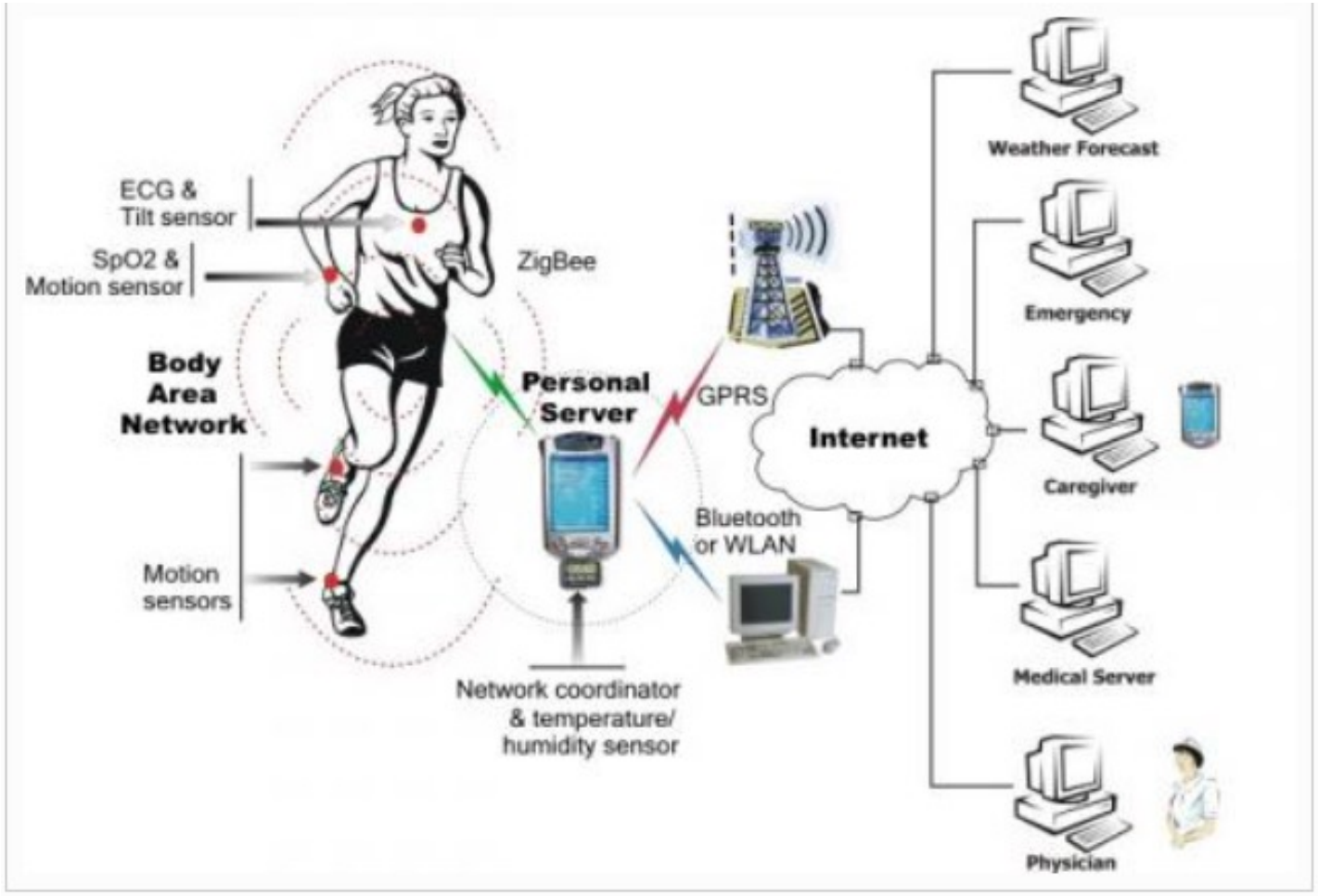
A medical device can be attached to the body with a small adhesive patch, implanted, or worn under clothing. This allows continuous activity monitoring and physiological and environmental measurements in a person's natural surroundings over extended periods.

There are several types of physiological medical devices:

1. Pills containing a wireless transceiver and sensors can detect enzymes, nucleic acids, intestinal muscle contractions, intestinal acidity, pressure, and other parameters. They allow noninvasive monitoring of gastrointestinal diseases.
2. Wearables are portable sensors with a wireless transceiver worn on the surface of the body. Examples include smartwatches and finger-worn ring sensors for monitoring heart rate.
3. Implantable physiological devices operate inside a patient's body. Examples include glucose level sensors and actuators for injecting insulin.

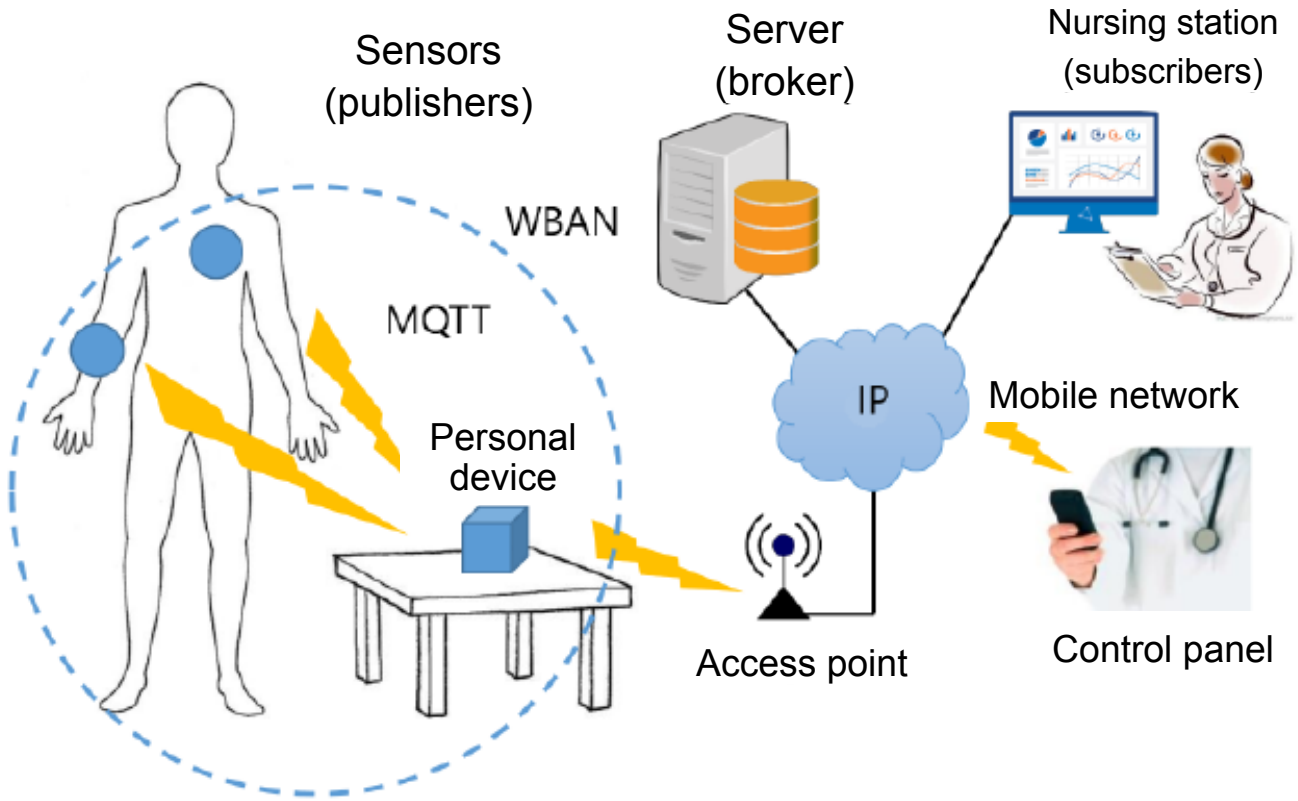
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WIRELESS BODY NETWORKS



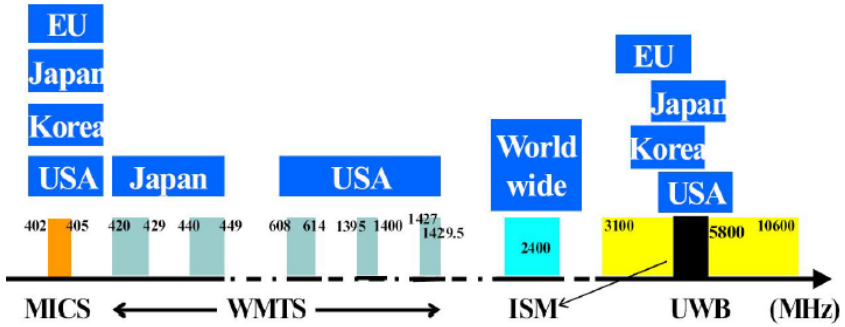
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WIRELESS BODY NETWORKS



WIRELESS BODY NETWORKS

- The Medical Implant Communication Service (MICS) band is a licensed band for implant communication. It uses the same frequency range (402–405 MHz) in most countries.
- Wireless Medical Telemetry Services (WMTS) is a licensed band for medical telemetry systems.
- The available bandwidth in both MICS and WMTS is insufficient for high-data-rate applications.
- The Industrial, Scientific, and Medical (ISM) band supports high-data-rate applications and is available worldwide. Interference is likely, however, because many wireless devices, including IEEE 802.1 and IEEE 802.15.4 devices, operate in the ISM band.



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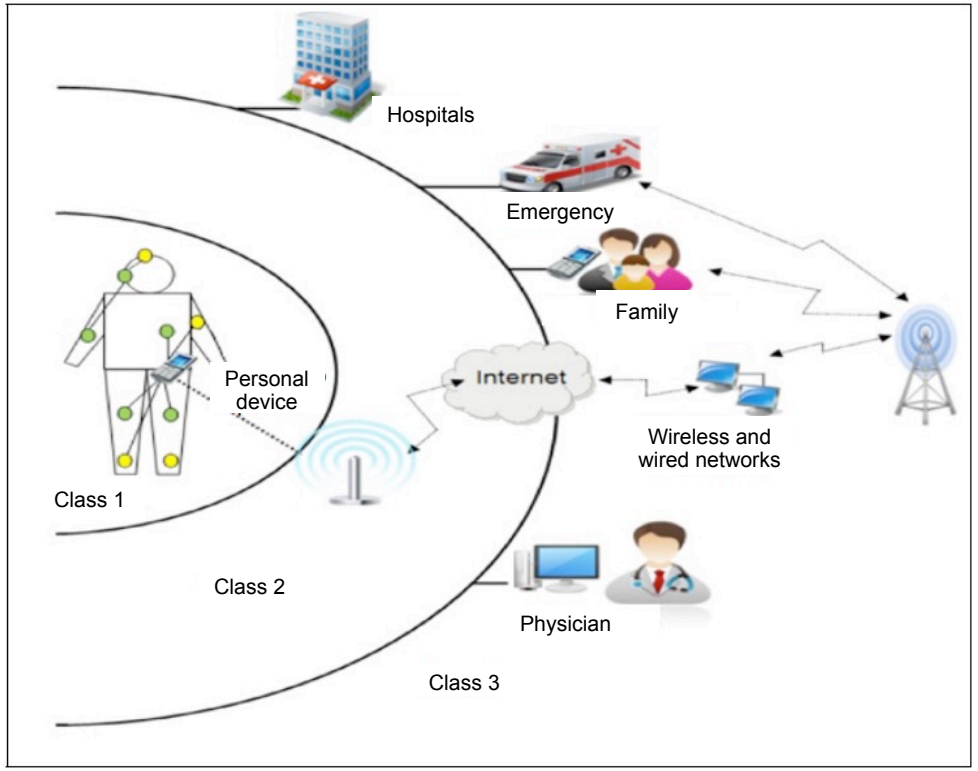
WIRELESS BODY NETWORKS

Frequency Band (MHz)	Number of Channels	Channel bandwidth	
402 – 405	10	300 kHz	Implantable
420 – 450	12	320 kHz	
863 – 870	14	400 kHz	
902 – 928	60	400 kHz	Wearable
950 – 958	16	400 kHz	
2360 – 2400	39	1 MHz	
2400 – 2483.5	79	1 MHz	

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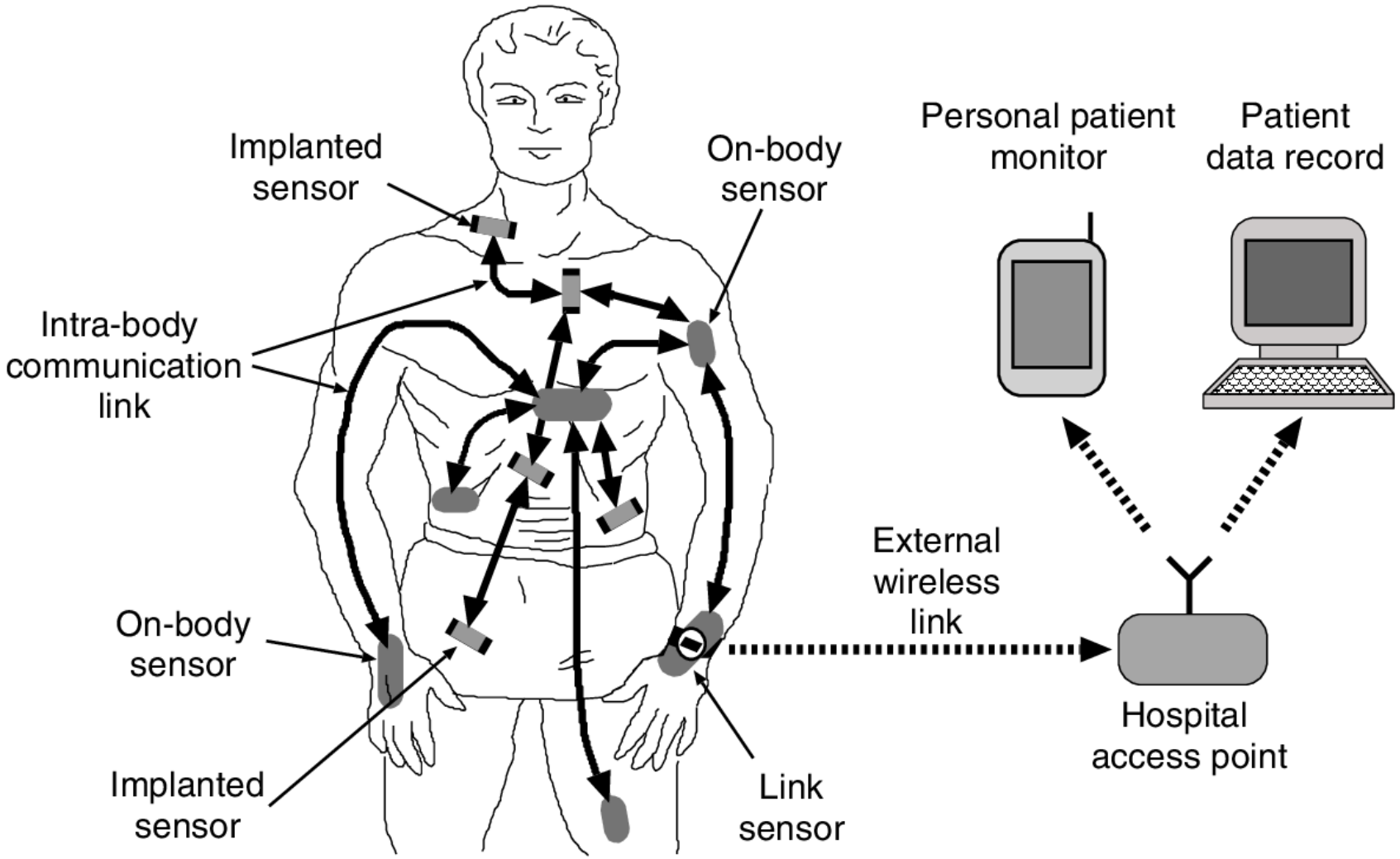
WIRELESS BODY NETWORKS

- WBAN communication can be divided into three classes:
  - Class 1: Intra-WBAN communication. Communication among the devices within a WBAN.
  - Class 2: Inter-WBAN communication. Communication between devices in different WBANs via their respective personal devices.
  - Class 3: Beyond-WBAN communication. Communication between the personal device and external networks, such as Wi-Fi or cellular networks.



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Wireless Body Networks



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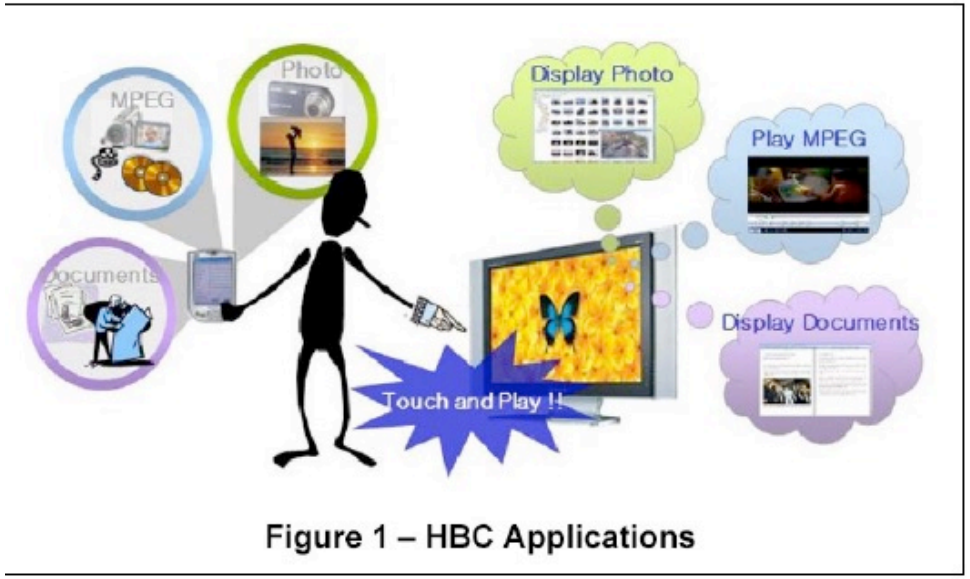


Figure 1 – HBC Applications

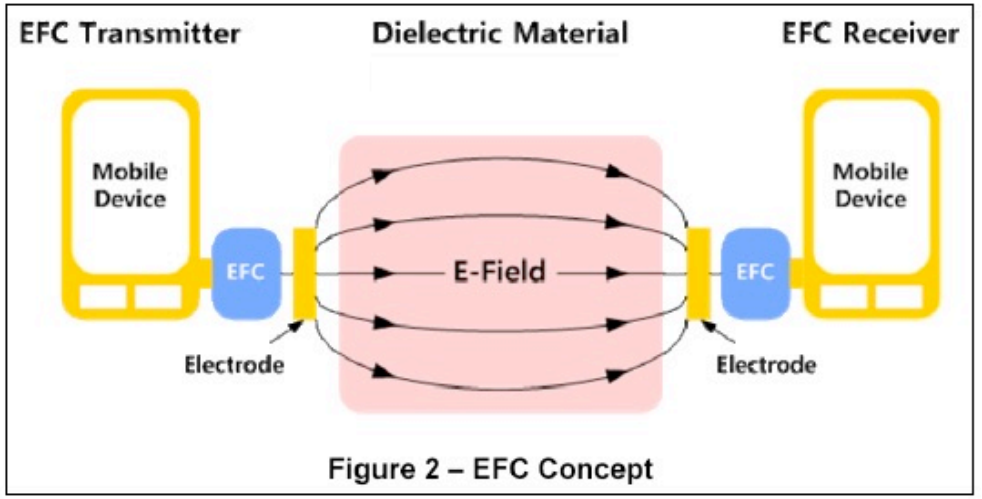


Figure 2 – EFC Concept

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Wireless Body Networks

- There are two main ways to couple signals to the body for intrabody communication (IBC): capacitive and galvanic coupling. Less common approaches include inductive coupling and conventional RF coupling.
- Capacitive coupling. Applying a signal to a transceiver electrode creates an electric field. This approach primarily aims to maximize coupling between the transceiver and the human body, reducing interference from ambient noise.
- Galvanic coupling. The flow of the applied current controls the signal, and the body can be treated as a waveguide. In contrast to capacitive coupling, galvanic coupling requires no return path for current propagation through the body. This makes it less susceptible to noise.

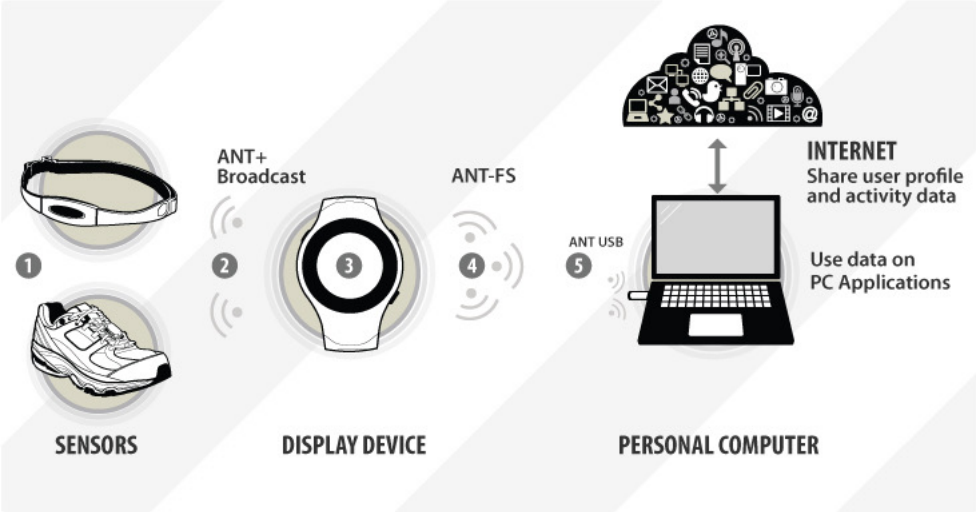
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Wireless Body Networks

- Inductive coupling. Electromagnetic coupling provides a communication link for implanted devices. An external coil near the patient couples with an implanted coil beneath the skin, and the coupled magnetic field powers the implant. Large transmit and receive coils provide the best power transfer, so this method works well when space is available.
- RF-based coupling is an alternative that can increase bandwidth and enable bidirectional data communication.

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PATHWAYS THAT CONNECT US TO THE FUTURE

ANT AND ANT+



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THE FUTURE

ANT AND ANT+

- Origin. A communication protocol was developed for a pedometer and a smartwatch to exchange distance and speed data during a run.



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PATHWAYS THAT CONNECT US TO THE FUTURE

ANT AND ANT+

- ANT is a protocol designed for wireless personal area networks (WPANs) with:
  - Low power consumption.
  - Low processing requirements.
  - Low transmission rates.
- It maximizes battery life and supports the use of many channels.

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PATHWAYS THAT CONNECT US TO THE FUTURE

ANT AND ANT+

- The compact size of ANT protocol stacks makes it possible to integrate them into devices such as PDAs, watches, phones, and smartphones.
- ANT operates in the 2.4 GHz ISM (industrial, scientific, and medical) band. It supports a bit rate of 1 Mbps and a transmission time of 150 microseconds per frame or less for 8 bytes of data.

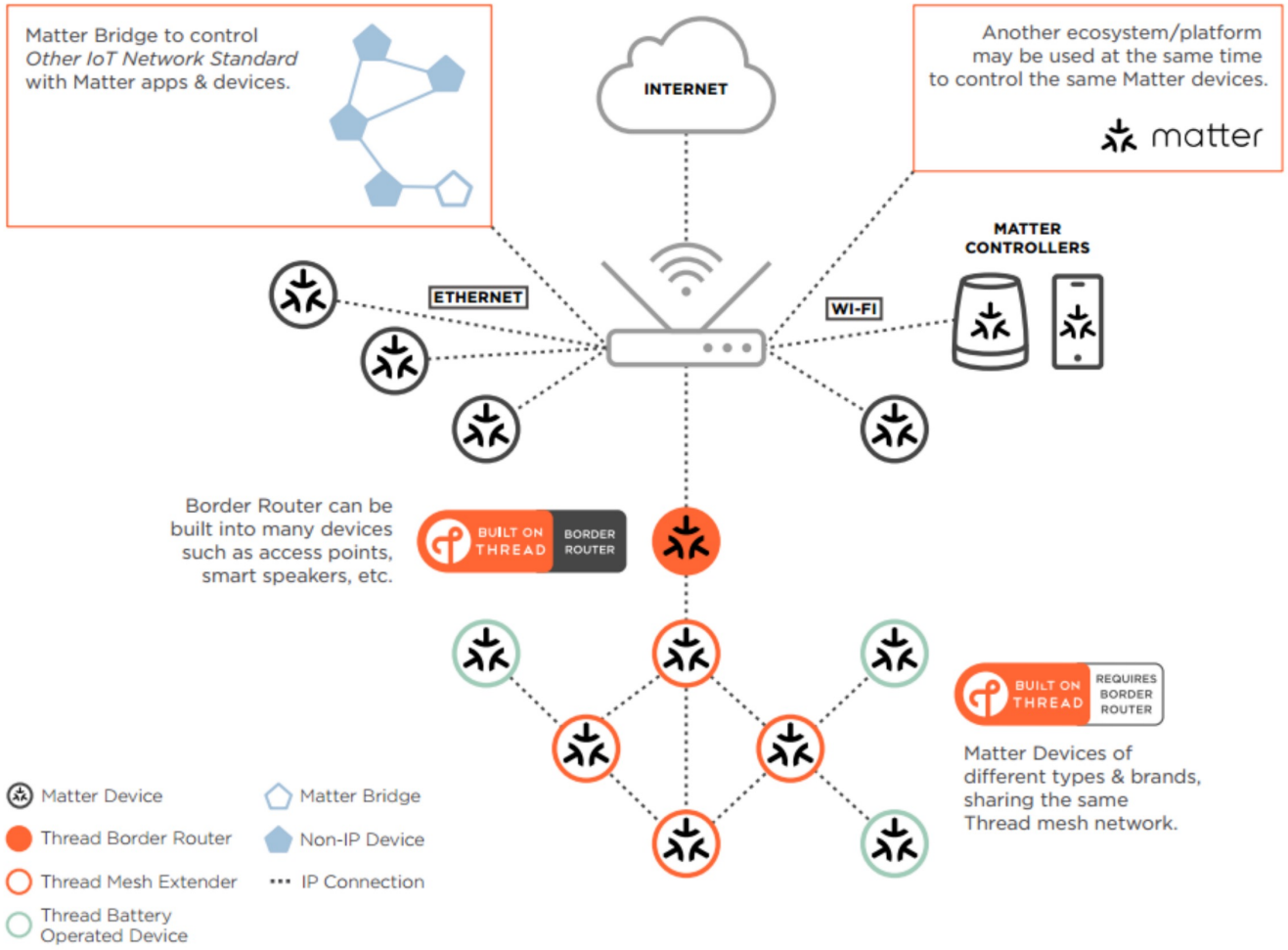
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PATHWAYS THAT CONNECT US TO THE FUTURE

ANT AND ANT+

- ANT (Advanced and Adaptive Network Technology) is a proprietary wireless sensor network technology. Its wireless communication protocol stack allows medical and industrial devices to communicate.
- ANT Wireless designs and markets ANT. ANT Wireless is a division of Dynastream Innovations Inc., a Garmin Inc. subsidiary. The ANT protocol is available in low-power RF transceivers from companies such as Nordic Semiconductor and Texas Instruments.
- ANT is a closed-source protocol with no standard or RFC.
- Companies such as Adidas, Garmin, Geonaute, Nike, Suunto, Fitbit, and Tacx use ANT to monitor devices.

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD


Matter Bridge to control
Other IoT Network Standard
with Matter apps & devices.
INTERNET
Another ecosystem/platform
may be used at the same time
to control the same Matter devices.
matter
MATTER
CONTROLLERS
ETHERNET
WI-FI
Border Router can be
built into many devices
such as access points,
smart speakers, etc.
BUILT ON THREAD
BORDER ROUTER
BUILT ON THREAD
REQUIRES BORDER ROUTER
Matter Devices of
different types & brands,
sharing the same
Thread mesh network.
Matter Device
Thread Border Router
Thread Mesh Extender
Thread Battery
Operated Device
Matter Bridge
Non-IP Device
IP Connection

THREAD

- Seven companies founded Thread Group, Inc.: ARM (SoftBank), Big Ass Fans, Freescale (NXP), Nest Labs (Google), Samsung, Silicon Labs, and Yale Locks.
- Thread Group, Inc. published version 1.0 of the Thread specification in 2015.
- Thread Group, Inc. currently has more than 200 members worldwide.



THREAD

- Thread is designed specifically for home automation in the IoT. It supports a wide range of home applications, including appliances, access control, temperature control, energy management, lighting, security, and safety.



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PATHWAYS THAT CONNECT US TO THE FUTURE

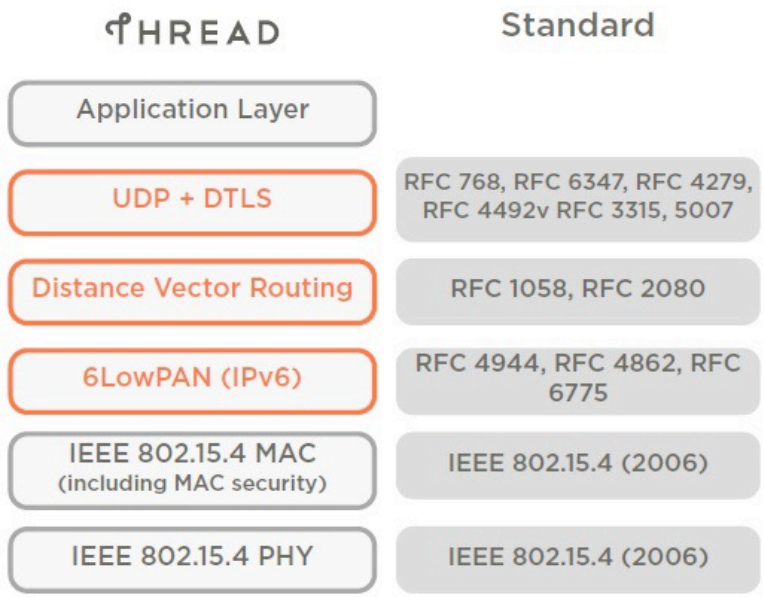
THREAD

- Thread is a low-power wireless radio protocol for smart homes that lets devices communicate directly with one another.
- In contrast to Zigbee and Z-Wave, Thread creates a mesh network by making each device in its network an access point.
- This distributes communication responsibilities more evenly, so each device does not need to communicate with a hub-like access point.
- Each Thread-enabled device can essentially act as a mini-hub.

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PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- Thread is intended to be an open standard and incorporates many existing and proposed standards.
- The Thread stack is an open standard based on a collection of standards from the Institute of Electrical and Electronics Engineers (IEEE) and the Internet Engineering Task Force (IETF).
- Thread is a secure wireless mesh networking protocol.



Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- General characteristics of the Thread stack:
  - **Simple network setup, startup, and operation.** Thread supports multiple network topologies. Setup is straightforward with a smartphone, tablet, or computer. Product installation codes ensure that only authorized devices can join. Simple protocols for forming and joining networks let systems configure themselves and correct routing problems as they occur.
  - **Security.** Only authorized devices can join, and all communications are encrypted and secured. The network layer provides security, and the application layer can also provide it. All Thread networks use Advanced Encryption Standard (AES) authentication and encryption.

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PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- General characteristics of the Thread stack:
  - **Large and small home networks.** Home networks range from a few devices to hundreds. The network layer is designed to optimize operation for the expected use.
  - **Large commercial networks.** In large commercial installations, a single Thread network cannot cover the full application or meet all system and network requirements. The Thread Domain model scales to 10,000 Thread devices in one deployment by combining technologies such as Thread, Ethernet, and Wi-Fi.
  - **Range.** Typical devices provide enough range to cover a standard home. Readily available designs with power amplifiers can substantially increase or vary the range. The physical layer (PHY) uses spread spectrum to improve interference immunity.

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PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- In commercial installations, the Thread Domain model lets multiple Thread networks communicate over a backbone. This extends coverage across many mesh subnets.
- **No single point of failure.** The Thread stack is designed to operate securely and reliably even when individual devices fail or are lost.
- **Low power consumption.** Devices communicate efficiently to improve the user experience and deliver years of expected battery life under normal conditions. With suitable duty cycles, devices can normally run for several years on AA batteries.
- **Cost-effectiveness.** Compliant chipsets and software stacks from multiple vendors are priced for mass deployment and designed from the outset for extremely low power consumption.

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PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- No single point of failure
- In a system of Thread devices, none of the devices represents a single point of failure. Some devices perform special functions, but the Thread network can automatically replace them without disrupting ongoing communication.
- For example, a Sleepy End Device (SED) needs a parent for communication, making that parent a single point of failure. If its current parent becomes unavailable, the SED can and will select another parent, so the user should not notice the failure.

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PATHWAYS THAT CONNECT US TO THE FUTURE

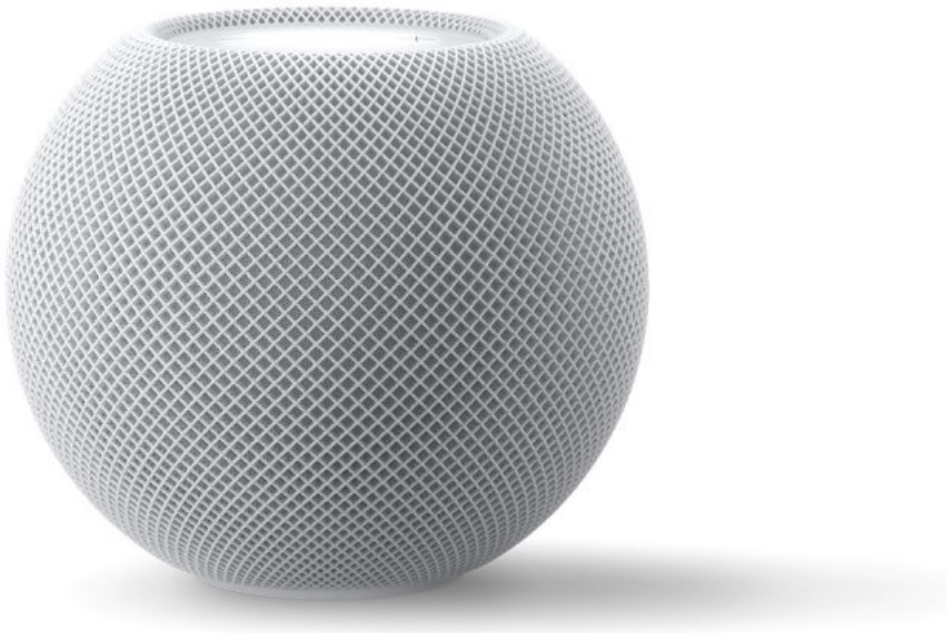
THREAD

- No single point of failure
- Thread is designed to avoid a single point of failure, although some topologies include individual devices without a backup. For example, in a system with one Border Router, there is no alternative if it loses power.
- One Router assumes the Leader role for certain Thread network functions. The Leader makes decisions for the network. For example, it assigns Router addresses and authorizes new Router requests.
- The network elects its Leader dynamically. If the Leader fails, another Router takes over. This autonomous operation avoids a single point of failure.

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PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- Devices:
- Apple HomePod Mini



Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- Devices:
- Google Nest Hub



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PATHWAYS THAT CONNECT US TO THE FUTURE

THREAD

- Devices:
- Eve Energy



THREAD

- Devices:
- Eve Flare



THREAD

- Devices:
- Eve Door & Window



THREAD

- Devices:
- Eve Weather



Works with Apple HomeKit

BUILT ON THREAD

Inatel

PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Z-Wave
- History
- Late 1990s: Zen-Sys, a Danish company, invented Z-Wave.
- The company originally set out to develop its own home automation solution and soon became a specialized supplier of home automation chips.
- An ecosystem of manufacturers with compatible products emerged.

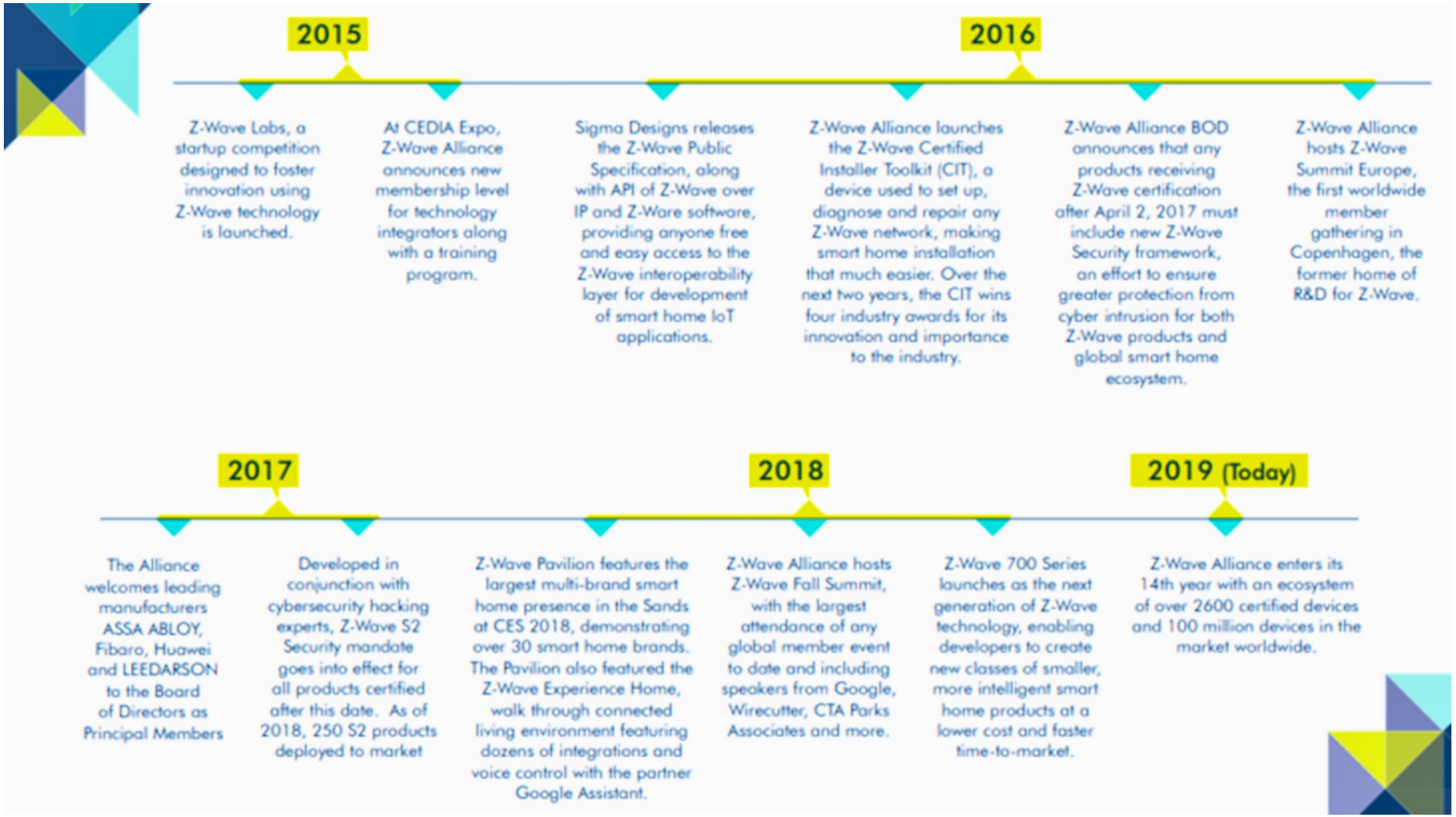
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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Z-Wave
- History
- The first generation of Zensys hardware went on sale in 2003.
- Four generations: 2003, 2005, 2007, and 2009.
- 2007: The first major Z-Wave device manufacturer in Europe was the German switch manufacturer Merten, now part of Schneider Electric.
- 2008: Zen-Sys acquired Sigma Design, an Asian manufacturer.
- 2009: Z-Wave gained adopters in Asia.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave


2015
Z-Wave Labs, a startup competition designed to foster innovation using Z-Wave technology is launched.
At CEDIA Expo, Z-Wave Alliance announces new membership level for technology integrators along with a training program.
Sigma Designs releases the Z-Wave Public Specification, along with API of Z-Wave over IP and Z-Ware software, providing anyone free and easy access to the Z-Wave interoperability layer for development of smart home IoT applications.
2016
Z-Wave Alliance launches the Z-Wave Certified Installer Toolkit (CIT), a device used to set up, diagnose and repair any Z-Wave network, making smart home installation that much easier. Over the next two years, the CIT wins four industry awards for its innovation and importance to the industry.
Z-Wave Alliance BOD announces that any products receiving Z-Wave certification after April 2, 2017 must include new Z-Wave Security framework, an effort to ensure greater protection from cyber intrusion for both Z-Wave products and global smart home ecosystem.
Z-Wave Alliance hosts Z-Wave Summit Europe, the first worldwide member gathering in Copenhagen, the former home of R&D for Z-Wave.
2017
The Alliance welcomes leading manufacturers ASSA ABLOY, Fibaro, Huawei and LEEDARSON to the Board of Directors as Principal Members
Developed in conjunction with cybersecurity hacking experts, Z-Wave S2 Security mandate goes into effect for all products certified after this date. As of 2018, 250 S2 products deployed to market
2018
Z-Wave Pavilion features the largest multi-brand smart home presence in the Sands at CES 2018, demonstrating over 30 smart home brands. The Pavilion also featured the Z-Wave Experience Home, walk through connected living environment featuring dozens of integrations and voice control with the partner Google Assistant.
Z-Wave Alliance hosts Z-Wave Fall Summit, with the largest attendance of any global member event to date and including speakers from Google, Wirecutter, CTA Parks Associates and more.
Z-Wave 700 Series launches as the next generation of Z-Wave technology, enabling developers to create new classes of smaller, more intelligent smart home products at a lower cost and faster time-to-market.
2019 (Today)
Z-Wave Alliance enters its 14th year with an ecosystem of over 2600 certified devices and 100 million devices in the market worldwide.

Z-wave

- Z-Wave alliance



Z-wave

- Z-Wave is a new wireless communication protocol for home automation devices.
- It is used in smart home applications such as lighting, security, and entertainment. It is one of the newer communication standards in the Internet of Things (IoT).
- Home automation interconnects electrically powered functions, such as lighting, heating, cooking, refrigeration, and security, and allows them to be automated.
- More than 300 manufacturers and thousands of devices currently use it, making it one of the most popular communication protocols after Zigbee.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave



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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- It works with a range of home products, including lights, thermostats, security sensors, and locks.
- It operates at a different frequency, avoiding interference with Wi-Fi or other signals in the home.
- A Z-Wave network can control up to 232 devices configured on that network.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

Z-Wave technology is currently widely used in home automation systems.

Z-Wave can also be used for efficient energy management and security applications such as power theft prevention.

Interoperability among Z-Wave devices makes it easier to combine applications such as lighting, smart security, and entertainment automation.

It is also used in industrial automation to keep interdependent processes running smoothly.

It is widely used in smart home security systems.

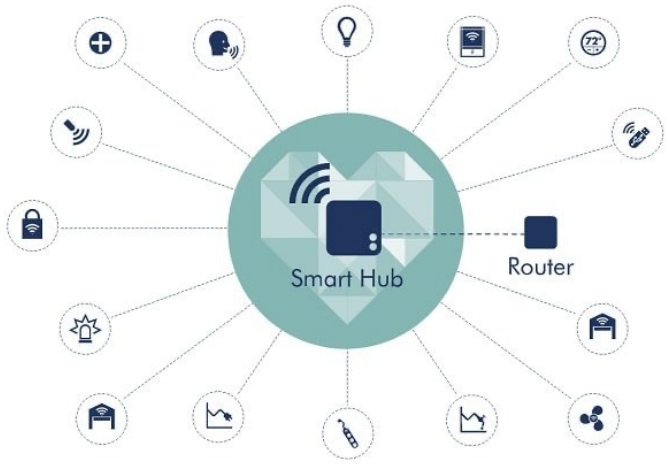
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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

Smart hubs are the central control units for all smart home applications.

Smart hubs can connect and configure a wide range of appliances, devices, power plugs, locking units, security systems, access control units, routers, sensors, and voice-enabled applications.

Smart hubs connect to the Internet through Ethernet or a wireless LAN. An Internet connection lets users access all connected devices from anywhere.



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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

Smart lighting is one of the most common applications of smart home technologies such as Z-Wave.

Light intensity can be set and controlled wirelessly from anywhere using a mobile app.

With a simple browser setup, users can add devices, configure functions, and schedule specific tasks for particular devices.

Smart lighting can help save energy through efficient control.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Smart locks are becoming popular. Z-Wave offers a wide range of smart access control solutions with easy-to-use keypads and touch panels.
- Z-Wave can be used to configure fingerprint scanners and highly secure access modes.
- Users can easily be added to the database, and selected members can be granted special access privileges.



Z-wave

- Smart sensors are essential components of smart homes and Internet of Things applications.
- Z-Wave smart sensors include thermostats, humidity sensors, proximity sensors, RFID tags, motion detectors, water level and flood sensors, smoke and gas sensors, and others.
- Smart sensor technology continues to evolve. Newer sensors are compact and energy-efficient, providing long battery life.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Z-Wave offers advanced home automation solutions.
- Any appliance compatible with Z-Wave can be added to a smart home network. Smartphone apps can control these devices and appliances securely.
- Z-Wave home automation solutions include smart plugs, smart outlets, power modules, temperature and smoke alarms, motorized smart window coverings, smart energy meters, and lighting controls.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Z-Wave offers smart solutions for indoor and outdoor security, monitoring systems, and alarms.
- Temperature and smoke detectors monitor continuously and trigger alarms when the measured level exceeds the preset threshold.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Voice-enabled applications are among the most advanced home automation applications.
- Voice commands can activate and control devices for specific tasks.
- This makes devices and applications more convenient to control.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Water management
- Water management systems and smart cities use accurate water level and flood sensors. Smart sensors monitor the water level continuously and trigger an alert when it exceeds the threshold.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Smart energy management
- Smart energy management is an important area of home automation. Energy management supports applications such as smart power modules, energy meters, and smart plugs for appliances.
- Smart lighting can save substantial energy through efficient use. It can be set to switch off or dim when a person leaves a room. Smart power modules can wirelessly control air conditioners, smart windows, heaters, fans, and other equipment.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Smart USB
- Smart USB turns any personal computer running Z-Wave software into a smart hub for home automation applications.
- Compatible devices can be easily connected to the network and controlled.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Devices:
- SAMSUNG SmartThings Smart Home Hub



Z-wave

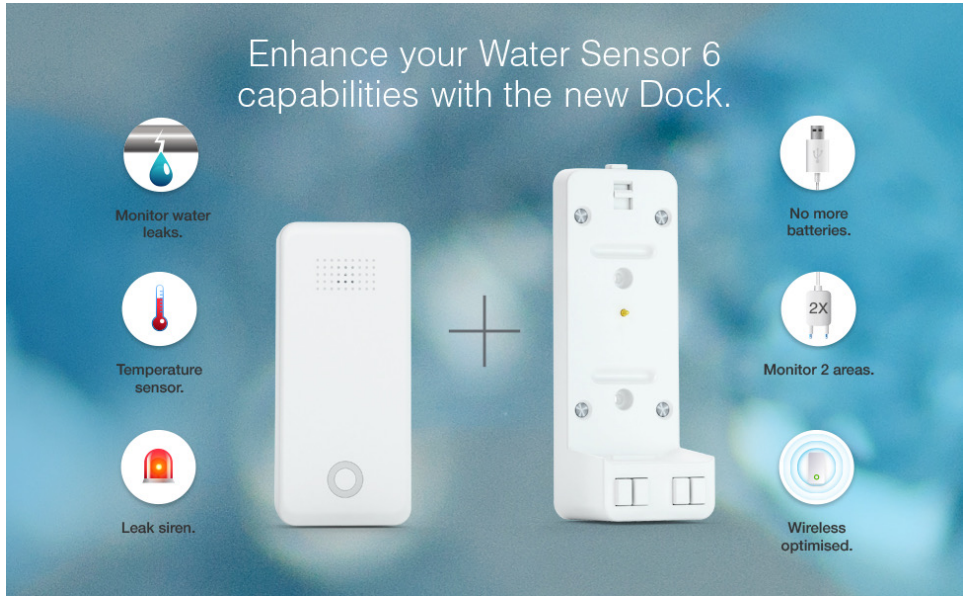
- Devices:
- Honeywell TH6320ZW2003 T6 Pro Series



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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Devices:
- Aeotec Dock for Z-Wave Water Sensor 6



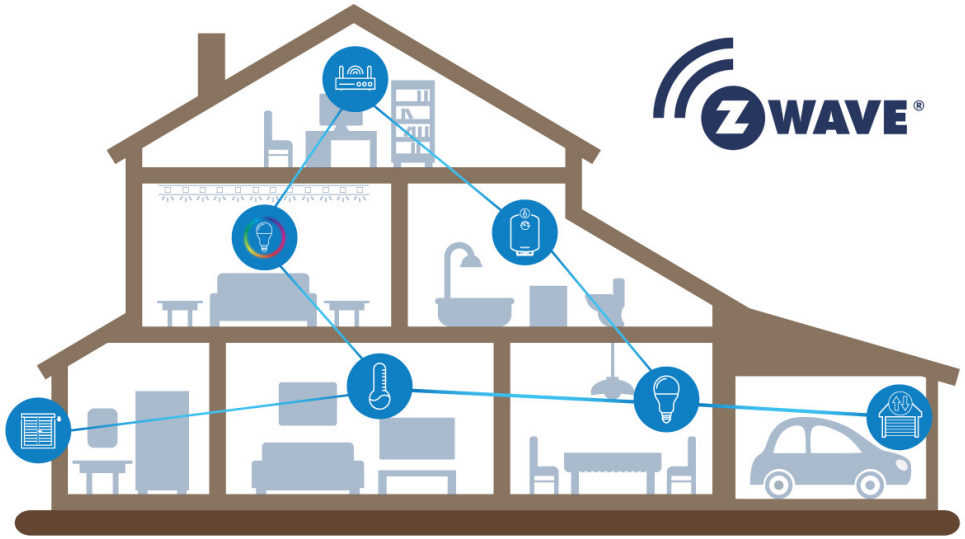
Z-wave

- Devices:
- Aeotec Home Energy Meter Gen5



Z-wave

- Z-Wave uses low-power radio waves.
- 900 MHz frequency bands.
- The mesh network covers all areas of the home.



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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Spectrum restrictions differ by location. Z-Wave is therefore region-specific, with different frequencies legally permitted in different regions.

COUNTRY / REGION	Z-WAVE FREQUENCY
South Africa	868.4 MHz
Australia	921.4 MHz
Brazil	921.4 MHz
Canada	908.4 MHz
China	868.4 MHz
United States	908.4 MHz
Hong Kong	919.8 MHz
India	865.2 MHz
Japan 920	922–926 MHz
Japan 950	951–956 MHz
Malaysia	868.4 MHz
Mexico	908.4 MHz
New Zealand	921.4 MHz
Russia	869.0 MHz
Singapore	868.4 MHz
European Union	868.4 MHz

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

- Z-Wave transmits at much lower power than Wi-Fi, giving users three to five years of battery life.
- Operating in the 900 MHz band results in less interference.
- Six layers of backward compatibility allow different versions to interoperate.
- Z-Wave provides 100 kbps data rates and enhanced security through AES-128 encryption.

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PATHWAYS THAT CONNECT US TO THE FUTURE

Z-wave

ZigBee	Z-Wave
Data Rate: 250kb/s	Data Rate: 40kb/s
Power Consumption: ~40mA	Power Consumption: ~2.5mA
Range: 10-20 meters	Range: 30-65 meters
Operates at 2.4GHz	Operates at 908MHz
Chips and modules available from multiple manufacturers	Chips only sold by Silicon Labs
Variable certification process	Strict certification process
Supports over 65,000 end nodes	Supports over 232 end nodes
More difficult to configure and set up	More user-friendly and easier to set up
More cost-effective	More expensive than Zigbee

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

enocean®



EnOcean

- What does “battery-free” mean?
- Which energy sources can be used?



EnOcean

- Based in Oberhaching, near Munich, the company provides valuable data for the Internet of Things (IoT) through resource-saving technology.
- For almost 20 years, EnOcean has made maintenance-free wireless switches and sensors that harvest energy from their surroundings, such as motion, light, or temperature.
- Miniaturized energy converters, ultra-low-power electronics, and robust radio technology based on open standards (EnOcean, Zigbee, and Bluetooth®) underpin digitized buildings, services, and production processes in the IoT.

EnOcean

- EnOcean offers a complete plug-and-play system of energy converters, power management, wireless modules, software, and development tools, together with a highly reliable, energy-saving radio protocol.
- All components of the platform are designed to work together seamlessly.
- Depending on energy requirements, the complete package supports a variety of wireless, battery-free applications.
- This comprehensive platform keeps integration barriers low. It is easy to integrate without deep specialist knowledge.

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

- Self-powered solutions are used in building automation, smart homes, LED lighting control, and industrial applications.
- EnOcean drives innovation as a partner to more than 350 leading product manufacturers. Its energy harvesting solutions have been deployed in more than a million buildings worldwide.

EnOcean


Batteryless Wireless Sensor
Optimization Algorithms
Control Software
Energy Converter
Micro Controller
Energy Management
Sensor
RF Transceiver
EnOcean core IP
Wireless Interface
enocean
zigbee
Bluetooth
Wireless System Module
RF Transceiver
Software
Micro Controller
Control Unit

EnOcean

- EnOcean Alliance - the wireless standard for sustainable buildings
- https://www.youtube.com/watch?v=YKYo6ns1i4o&t=3s



EnOcean



Switch applications

Battery-free wireless switches use kinetic energy converters for switching applications in buildings and the Internet of Things.

Battery-free sensors

Our wireless sensors harvest energy from ambient light using small solar cells.

Building automation

Self-powered sensors are used in building automation systems. They enable intelligent, energy-efficient control of lighting, heating, security, and other services. EnOcean offers a complete portfolio of wireless modules and development tools for edge devices, gateways, and distributed control systems.

Wireless LED control

For LED lighting control, we offer a complete package based on the 902 MHz EnOcean radio standard, with transmitters, receivers, and a commissioning tool.

IoT

We deliver sustainable, wireless, maintenance-free solutions for digitization. Our self-powered sensors integrate seamlessly with IoT platforms.

EnOcean

- EnOcean can enable virtually any ultra-low-power radio to operate on harvested energy.
- EnOcean technology currently powers the international ISO/IEC 14543-3-1X standard, also known as the EnOcean standard, which is optimized for ultra-low-power wireless applications and energy harvesting.

EnOcean

- EnOcean can enable virtually any ultra-low-power radio to operate on harvested energy.
- EnOcean technology currently powers the international ISO/IEC 14543-3-1X standard, also known as the EnOcean standard, which is optimized for ultra-low-power wireless applications and energy harvesting.

EnOcean

- Ultra-low-power subprotocols of existing wireless standards can use EnOcean technology. One example is the 2.4 GHz ISM band (IEEE 802.15.4).
- These specifications draw on features of the EnOcean standard to enable ultra-low-power applications in their established fields.
- Examples: Zigbee and Bluetooth.

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

- EnOcean recommends ISO/IEC 14543-3-1X for sustainable, reliable integrated building automation and intelligent control of interconnected building zones that require high system availability.
- The EnOcean radio protocol is designed specifically for ultra-low-power devices and energy harvesting applications in home and building automation.

EnOcean

Wireless standard for building automation and smart homes (ISO/IEC 14543.3.1x)

- Uses regulated frequency bands with greater airtime availability (approved for pulsed signals only).
  - 868 MHz under RED regulations.
  - 902 MHz under the FCC/IC specification.
  - 928 MHz under the ARIB specification.
- Multiple transmissions of telegrams.
- Data rate of 125 kbit/s.
- ASK and FSK modulation.
- Short telegrams (approximately 1 ms) reduce the chance of collisions, allowing many EnOcean sensors to operate in one system.
- Long range: up to 30 meters inside buildings and 300 meters in open terrain.

EnOcean

- For optimal RF performance, the radio protocol uses frequency bands below 1 GHz. It achieves reliable RF operation by sending wireless signals lasting less than 1 millisecond at 125 kbps.
- Transmit power can reach 10 mW, yet one wireless telegram requires only 50 µW·s of energy.
- That is about the energy needed to lift a 1-gram weight by 5 millimeters. The short telegram is repeated twice at random times within about 40 milliseconds to avoid transmission errors.

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

Wireless standard for building automation and smart homes (ISO/IEC 14543.3.1x)

- One-way and two-way communication.
- Enhanced data security through AES-128 encryption.
- No interference with other systems.
- System design verified in an industrial environment.

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PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

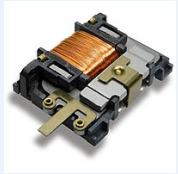
- A wireless protocol for low-power devices such as energy harvesters in the home. It is specifically designed to keep the power consumption of these sensors and switches extremely low.
- The design achieves this by:
  - Keeping communications very short, infrequent, and primarily one-way.
  - Using frequencies that provide good range even at low transmit power and avoid collisions.

EnOcean

- Small, inexpensive energy harvesters can compete with similar battery-powered devices.
- EnOcean modules combine micro-energy converters, ultra-low-power electronics, and reliable wireless communication. This lets EnOcean customers build self-powered wireless sensor solutions for collecting and processing data in a smart system.

ECO 200

Kinetic energy harvesting



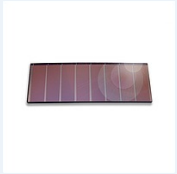
The kinetic converter and a wireless module enable many battery-free switching applications.

Energy from a switching action (a button press):

- Electrodynamic energy converter
- Generates power from motion
- Typically more than 1,000,000 switching cycles at 25 °C
- For small, flat switch designs

ECS 300

Solar cell for self-powered wireless sensors



Solar cell for energy-harvesting wireless sensors. ECS 300: 35.0 × 12.8 × 1.1 mm.

The smaller ECS 300 can be used for one-way sensor applications.

ECT 310 Perpetuum

Thermal energy harvesting



Thermal converter for energy-harvesting wireless sensors and actuators.

Heat dissipation as a source of energy:

- Standard Peltier element with an ultra-low-power EnOcean DC/DC converter
- Uses a minimal temperature difference
- Maintenance-free, fully integrated
- Enables energy-harvesting actuators

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

- Most energy harvesters supply only small amounts of energy. The system must accumulate energy over time while asleep and use only a small fraction of it in the process.
- EnOcean sensors therefore have extremely low idle current. The system consumes very little energy while asleep.
- The latest generation of EnOcean energy-harvesting wireless sensors requires standby currents of just 100 nanoamperes (nA) or less.

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean



Batteryless wireless switches control light and shading

Batteryless wireless window contacts monitor window status

Batteryless wireless switches perform remote-control functions

Batteryless wireless door/window handles monitor door/window status

Occupancy sensors trigger lighting and climate control systems

Batteryless wireless control units allow for optimal climate conditions and maximum operating comfort

Wireless connectors control and monitor household appliances

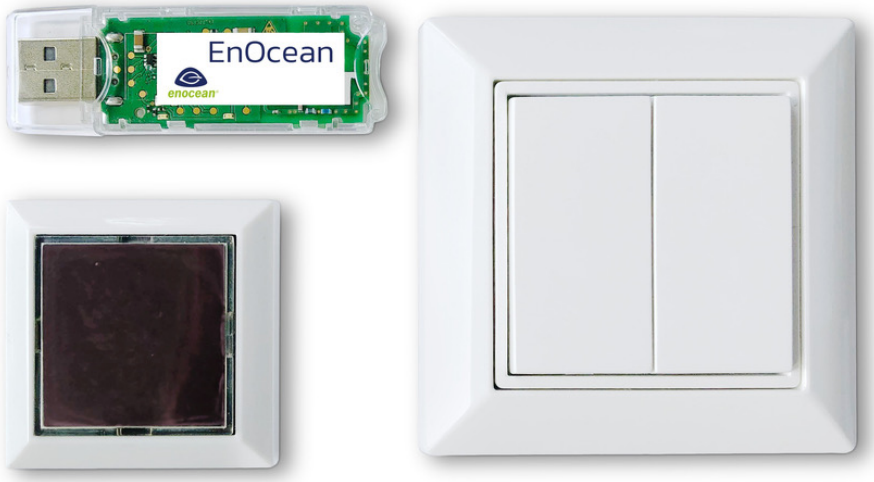
Wireless actuators control radiators, room controllers govern underfloor heating

Networked smoke sensors set off fire alarms to trigger emergency response

Sunblind actuators control the sun-shade elements

Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean



EnOcean

- Applications: Temperature sensor
- FTR65SB-wg wireless temperature controller.
- Power supply with integrated solar cell and battery (5-year service life).



Inatel
PATHWAYS THAT CONNECT US TO THE FUTURE

EnOcean

- Applications: Motion sensor
- The wall-mounted Easyclick wireless motion sensor is powered by solar cells and detects movement using a passive infrared sensor.
- The occupancy sensor is easy to install and saves energy. For example, it automatically switches the lights ON when someone enters a room and switches them OFF after a delay once the person leaves.



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EnOcean

- Applications: Atmospheric pressure sensor



EnOcean

- Applications: NodOn® Micro Smart Plug
- Switch any connected device ON or OFF. Its very small size makes this smart plug discreet and makes your home smarter with minimal effort.



SCHUKO TYPE E

EnOcean

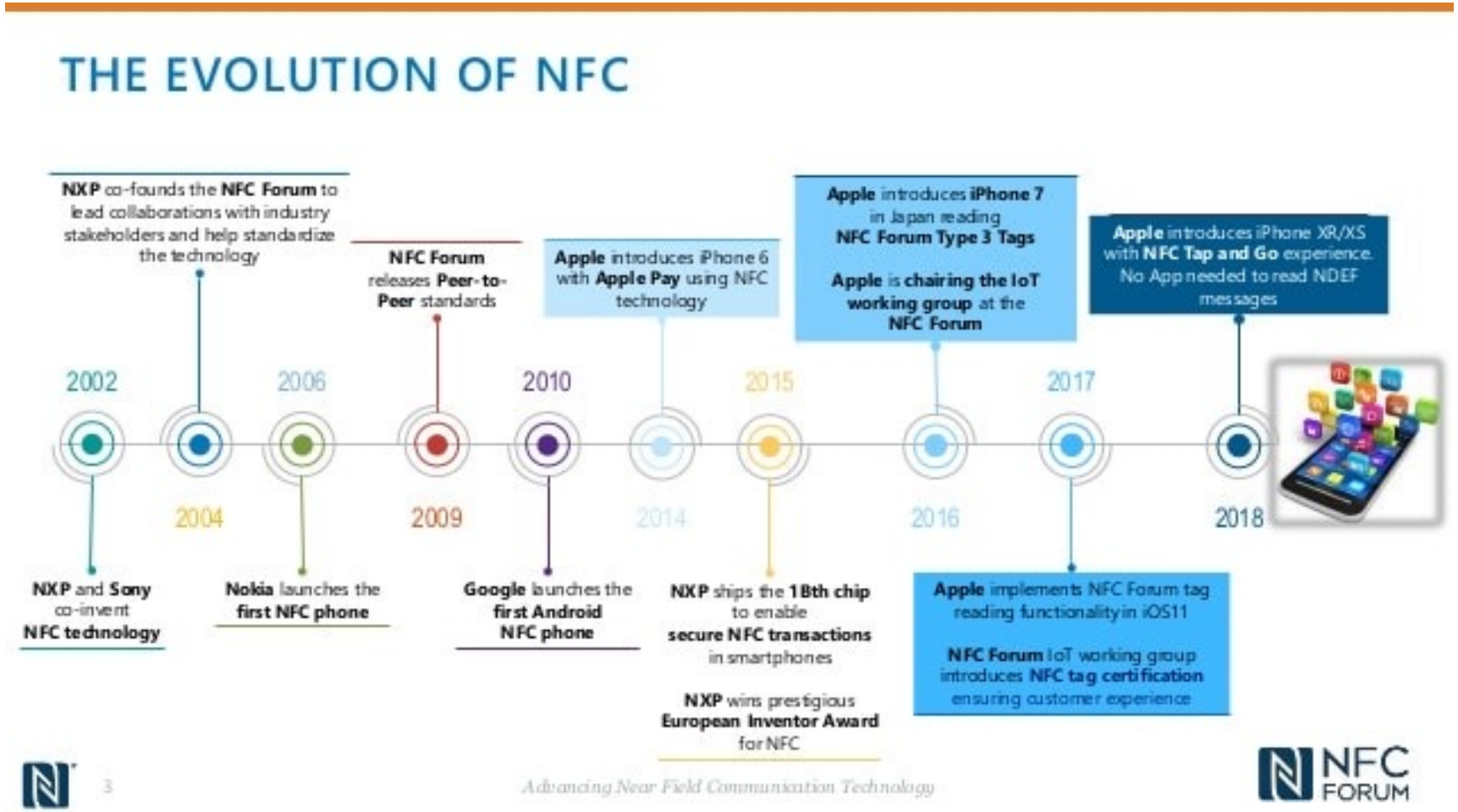
- EnOcean Wireless LED Controls
- LED lighting is one of today's most promising and fastest-growing technologies. EnOcean is a worldwide radio standard for self-powered switches and sensors.
- It provides maximum flexibility, comfort, and safety in a building. This proven technology is energy-efficient and saves both time and money.
- EnOcean now offers a comprehensive 902 MHz LED control system for the North American market. It combines innovative self-powered sensors and switches with LED fixture controllers and commissioning tools to simplify installation and setup.
- https://youtu.be/GFGN4H54RDw

NFC



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NFC

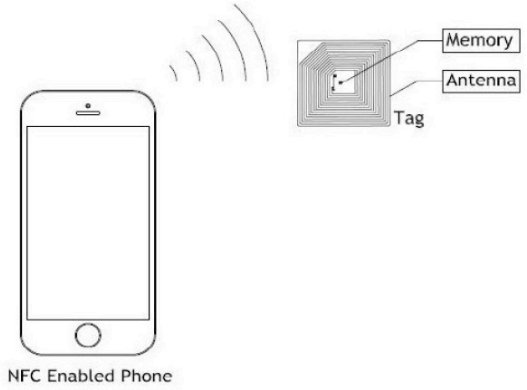

THE EVOLUTION OF NFC
2002
NXP and Sony co-invent NFC technology
2004
NXP co-founds the NFC Forum to lead collaborations with industry stakeholders and help standardize the technology
2006
Nokia launches the first NFC phone
2009
NFC Forum releases Peer-to-Peer standards
2010
Google launches the first Android NFC phone
2014
Apple introduces iPhone 6 with Apple Pay using NFC technology
2015
NXP ships the 1Bth chip to enable secure NFC transactions in smartphones
NXP wins prestigious European Inventor Award for NFC
2016
Apple introduces iPhone 7 in Japan reading NFC Forum Type 3 Tags
Apple is chairing the IoT working group at the NFC Forum
2017
Apple implements NFC Forum tag reading functionality in iOS11
NFC Forum IoT working group introduces NFC tag certification ensuring customer experience
2018
Apple introduces iPhone XR/XS with NFC Tap and Go experience. No App needed to read NDEF messages
3
Advancing Near Field Communication Technology
NFC FORUM

NFC

- Near-field communication (NFC) is based on a simple idea. Two nearby conductive coils can exchange electrical energy over short distances (<5 cm) through wireless inductive coupling.
- NFC emerged almost two decades ago as a secure, short-range wireless data exchange technology. Its ability to transfer energy and data at the same time offers opportunities to design miniature, battery-free, disposable sensing systems for healthcare and food quality monitoring.

NFC

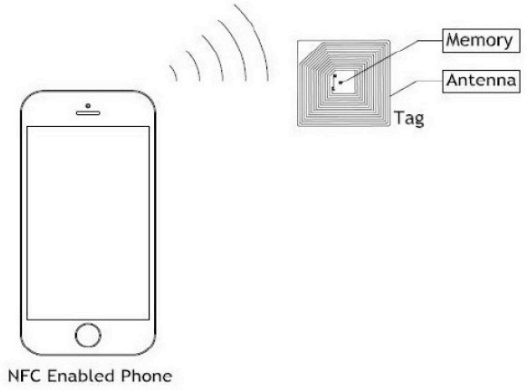
- NFC communication requires two devices.
- The first is the initiator, an active device that starts communication. The second is the target, which responds to the initiator's requests.
- The target can be active or passive. Communication starts when the active device approaches the target, generates a 13.56 MHz magnetic field, and powers the target.



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PATHWAYS THAT CONNECT US TO THE FUTURE

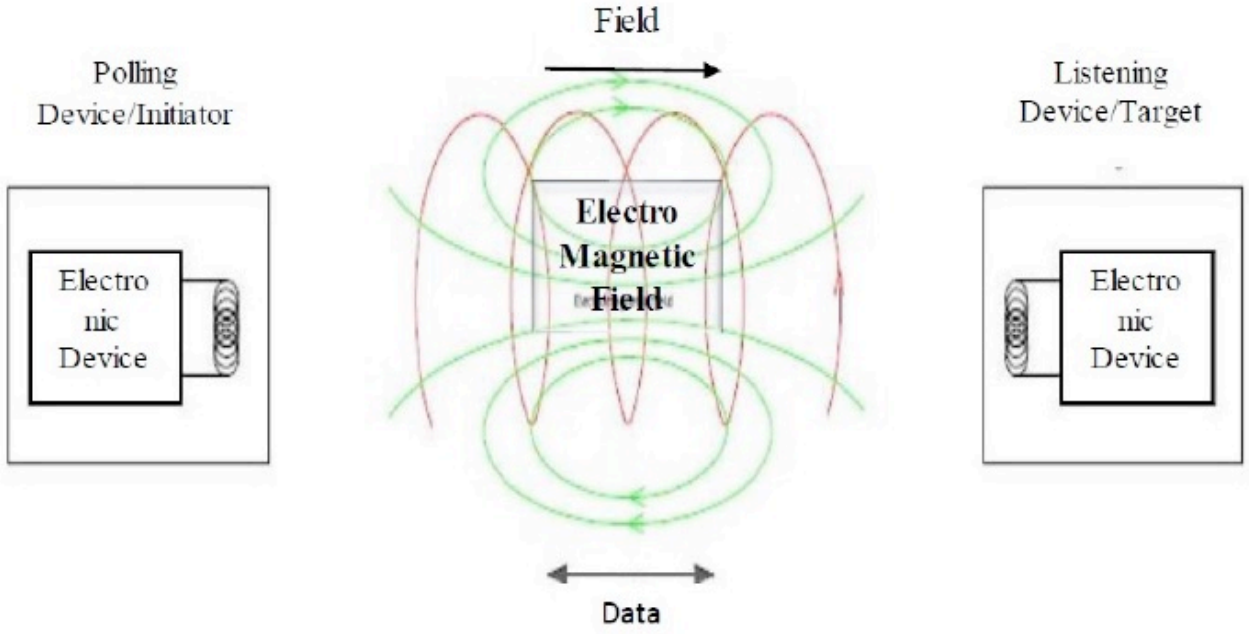
NFC

- NFC uses magnetic field induction and operates in an unlicensed radio frequency band.
- NFC operates at 13.56 MHz and can transmit data at up to 424 kbit/s.
- The initiator also includes a built-in power source. The target can be an RFID card, a tag, or an NFC device that responds to the initiator’s request.



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NFC



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NFC

- NFC defines two device types: the initiator and the target.
- The initiator starts communication and controls the data exchange.
- The target responds to the initiator. NFC has two operating modes, active and passive.

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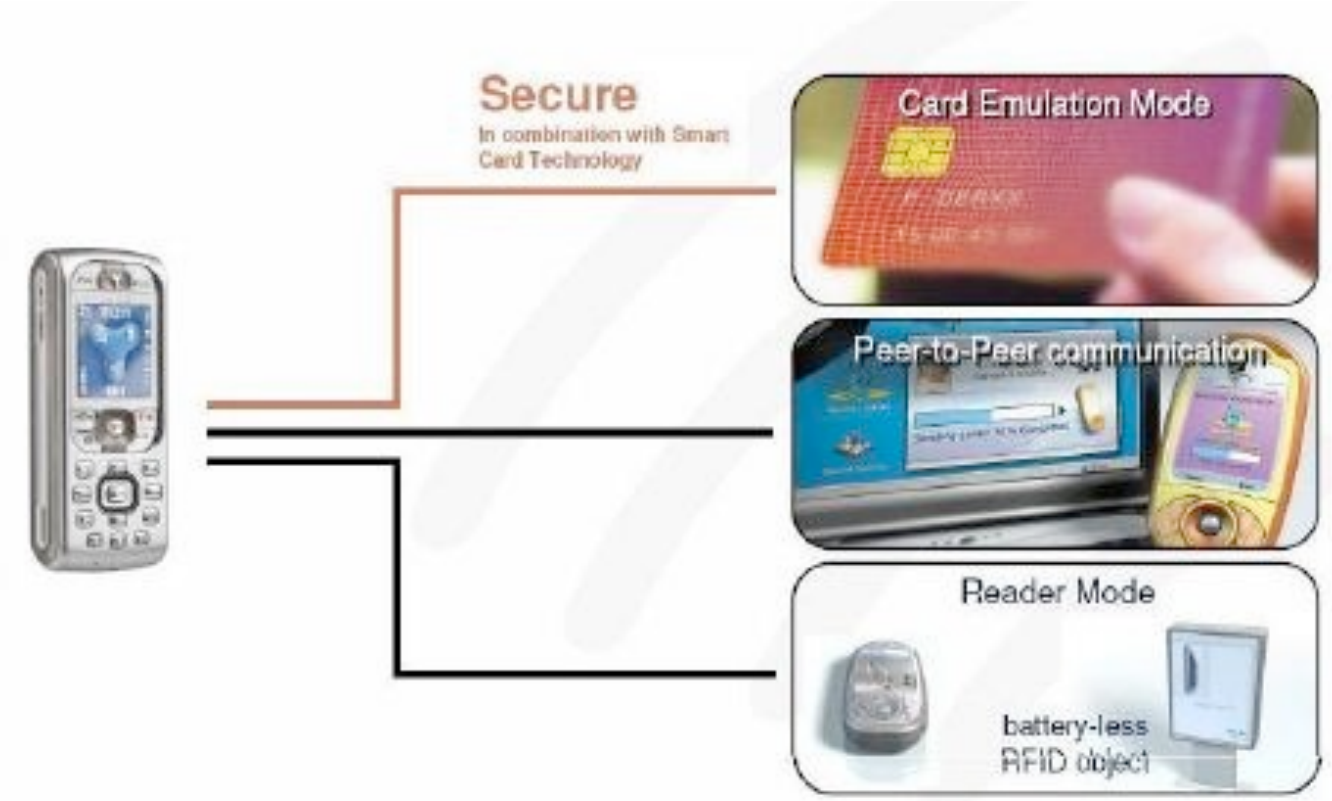
NFC

- Active and passive are the two NFC operating modes.
- In active mode, both the initiator and the target generate the RF signal that carries the data.
- In passive mode, only the initiator generates the RF signal. The target responds using a technique called load modulation.

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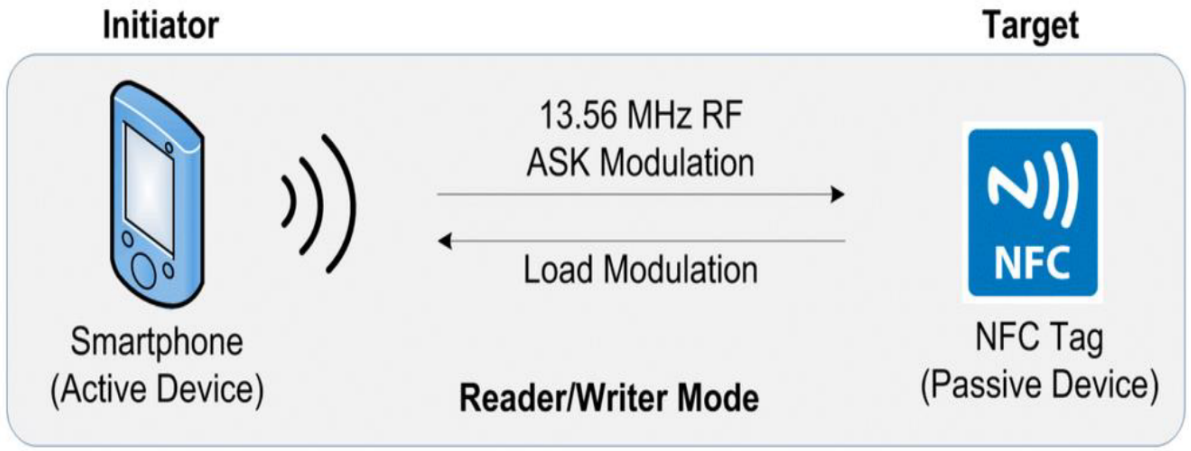
NFC

- In addition to the two operating modes, there are three modes for device communication.



NFC

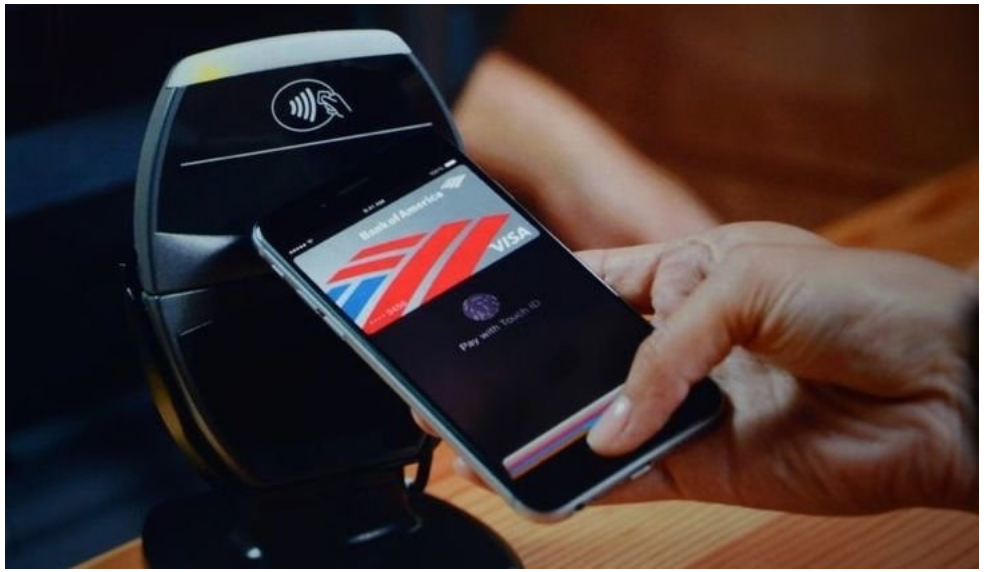
- In reader/writer mode, the application transfers data in the message format defined by the NFC Forum.
- In this mode, an NFC-enabled mobile phone can read and write NFC tags.
- In reader mode, the NFC initiator reads data from the NFC tag. In writer mode, it writes data to the tag. Reader/writer communication is insecure.



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NFC

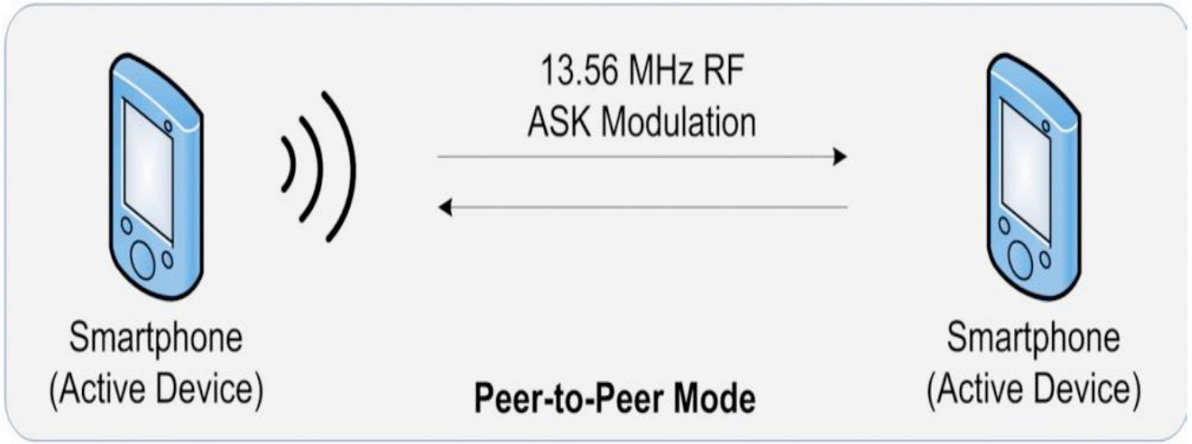
- In card emulation mode, an NFC-enabled mobile device acts as a contactless smart card.
- Smart cards include debit cards, credit cards, access cards, and others.
- Data transfer in this mode is highly secure.



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NFC

- Peer-to-peer mode supports link-level communication.
- It allows two NFC-enabled devices to exchange information such as a text message, a contact record, or any other type of data.
- Data transfer in this mode is highly secure.



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NFC

- Advantages of NFC for industrial applications include:
  - NFC allows easy, tap-based communication between two devices.
  - Setting up an NFC connection takes milliseconds, while Bluetooth typically takes seconds.
- NFC can extend sensor battery life in wireless sensing applications or even enable a battery-free sensor.

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PATHWAYS THAT CONNECT US TO THE FUTURE

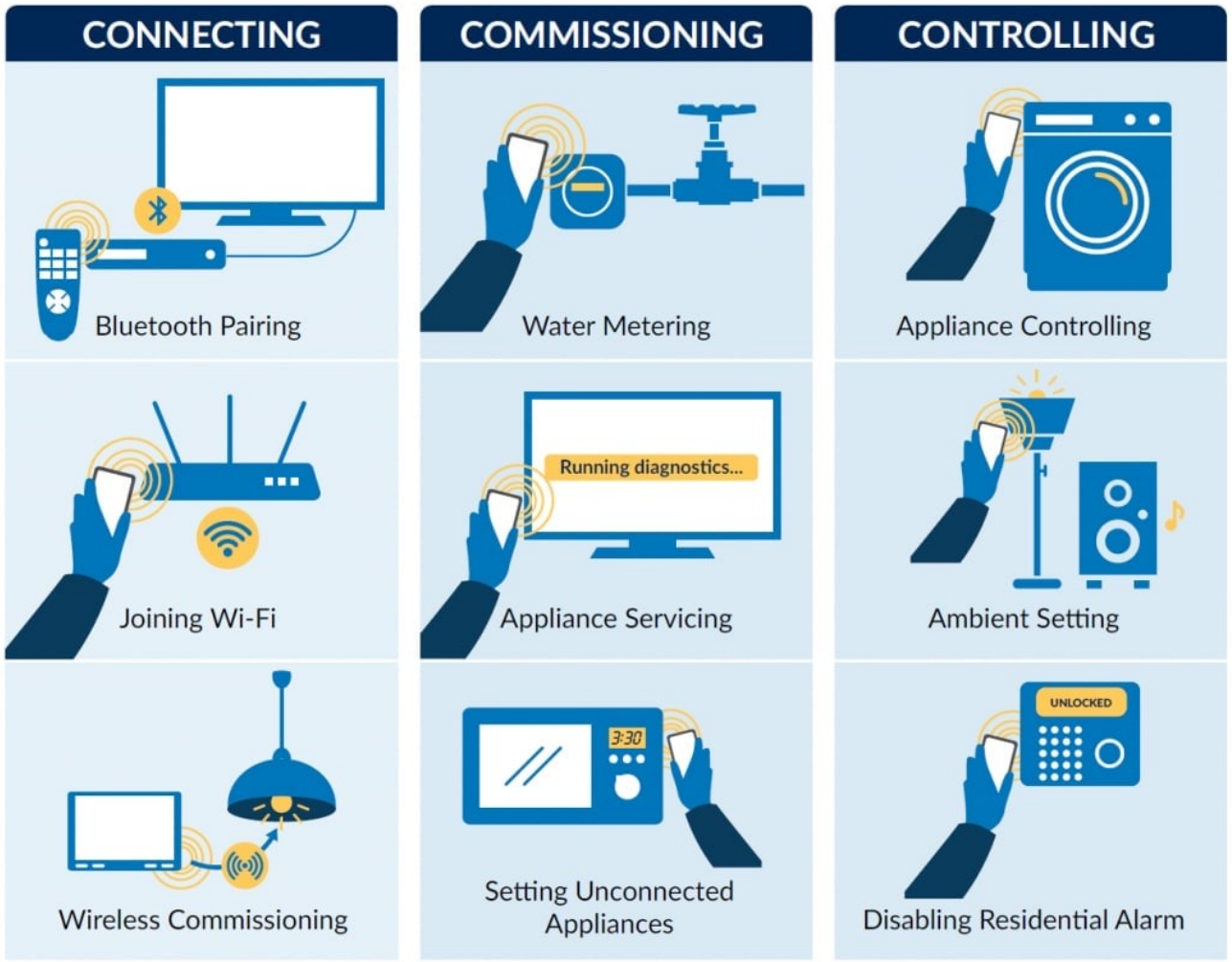
NFC

- **Mobile payment.** Customers with NFC-enabled smartphones can connect to their bank accounts and pay by tapping the phone on an NFC-enabled point-of-sale terminal.
- **Access and authentication.** Keyless access to restricted areas, cars, and other vehicles. Another possible future use is controlling household devices with NFC.
- **Transport and ticketing.** NFC-enabled phones can connect to an NFC kiosk to download a ticket, or the ticket can be sent directly to the phone over the air (OTA). The user can then tap the phone on a reader to redeem the ticket and gain access.
- **Mobile marketing.** NFC tags can be embedded in indoor and outdoor signage. By tapping an NFC-enabled smart poster with a smartphone, a customer can read a consumer review, visit a website, or even watch a movie trailer.

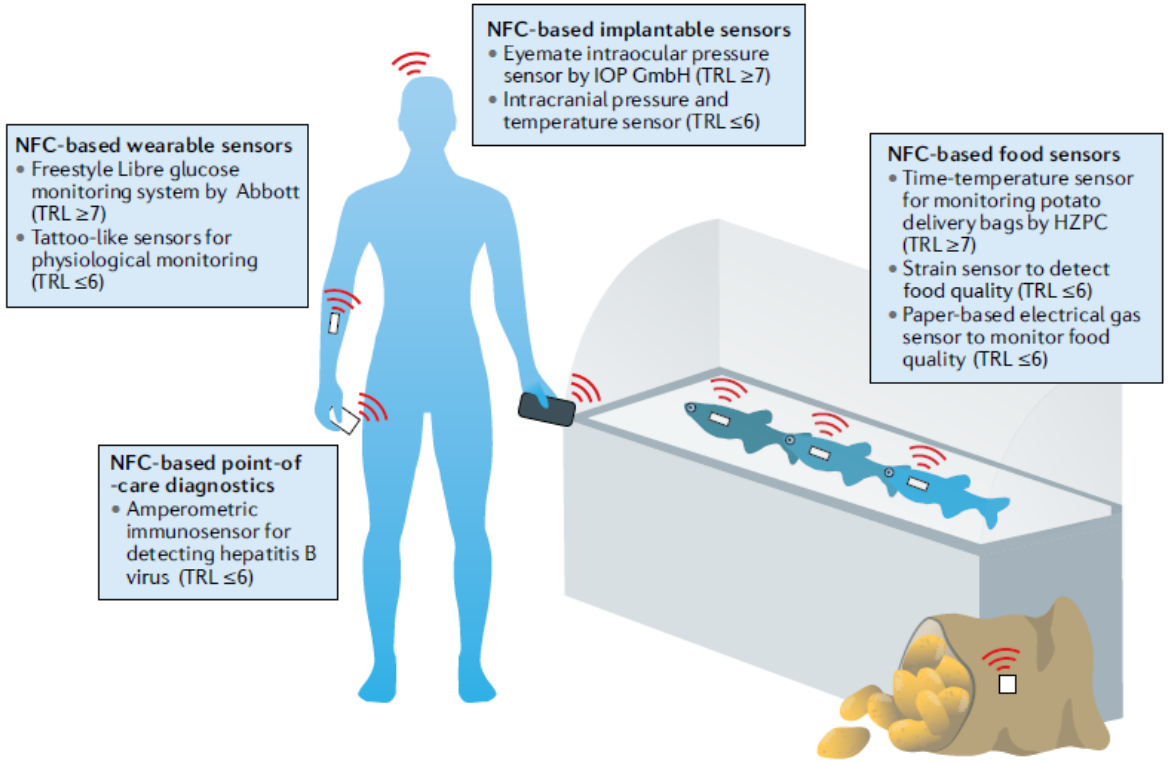
NFC

- **Healthcare.** NFC medical cards and wristband tags can store relevant, up-to-date patient information, including medical history, allergies, and infectious diseases.
- **Gaming.** NFC bridges physical and digital games. Players can connect their phones to one another and use the NFC app to earn bonus points, unlock a new level, or get clues.
- **Inventory tracking, smart packaging, and shelf labels.** NFC-tagged objects can provide a wide range of information in different settings.
  - Readers can tap tags with NFC-enabled smartphones to access book reviews and author information, and recommend books to others.
  - Users can check out a book or add it to a wish list to check out later.
  - With NFC, library records and metadata could theoretically be stored on and retrieved from the physical items themselves. A user could scan a borrowed book or resource to check its title, author, and due date.

NFC


CONNECTING
COMMISSIONING
CONTROLLING
Bluetooth Pairing
Water Metering
Appliance Controlling
Joining Wi-Fi
Running diagnostics...
Appliance Servicing
Ambient Setting
Wireless Commissioning
3:30
Setting Unconnected Appliances
UNLOCKED
Disabling Residential Alarm

NFC



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NFC

- Kraft Heinz uses packaging with near-field communication (NFC) technology to engage with customers in the store before they buy the product.
- A customer enters a store such as Walmart, picks up a pack of Kraft Singles cheese, and uses NFC to visit a designated website. There they can enter their details for a contest and access recipes, coupons, and other rewards.



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NFC

- The São Paulo city government began testing contactless payments on buses in September 2019.
- Only some routes have buses equipped with the technology. All vehicles that accept contactless card payments for the fare will display an identifying sticker.



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NFC

Contactless payment

Pilot project. Tap your card on the reader.

Advertise here.

ARTESP. São Paulo State Transport Agency. Oversight and control of toll roads. 0800 727 8377. Department of Transport.



NFC

- Social life
- At a pub in Sweden, people can share their check-in information with friends at the same venue. They choose to share by tapping an NFC-enabled phone on a location tag.
- Berlin, Germany, has done the same with NFC tags at several locations around the city. People can share their location with a tap. A single tap makes it quick and easy to share a check-in with friends.

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NFC

- The Museum of London installed NFC tags near artifacts and at various locations around the museum.
- Visitors can learn more about these artifacts by tapping the tags with their NFC-enabled phones. One simple action reveals information about physical objects and enriches the museum experience.
- NFC makes the boundary between the physical world, data, and people almost invisible.

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NFC

